

Application of Built-in Adaptive Remeshing and Mesh Refinement Features in Abaqus to Fracture Simulations Using Phase-field User Elements

Comprehensive Mode-I Convergence & UEL Energy Audit Supervisor Report – 08 October 2026

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1. Executive Summary and Governance Framework

Governing Supervisor Directive & Active Phase

“We need to have understood everything related to the first model before we increase complexity.”

MODE1_GATE6B_ACTIVE_EVALUATION_AND_CONTINUATION

Following the supervisor decisions of 17 September 2026, this report provides the comprehensive Mode-I convergence and energetic audit package for the 08 October 2026 supervisory review. The primary objective is to complete the physical, numerical, and energetic qualification of the Mode-I phase-field benchmark before advancing to state transfer, visualization integration, or multi-crack geometries.

1.1 Status of governing milestone gates

- Gate 6A – Mechanical Implementation & Defect Closure: RESOLVED_AND_CLOSED.** The early 71,320-element initial stiffness defect was proven to arise from an Abaqus keyword preprocessing line limit (≤ 16 nodes per line in *NSET without GENERATE), silently omitting 134 of 150 boundary nodes. Wrapping the data lines restored all constraints and achieved reference-consistent structural stiffness recovery (within 0.09%, $K_0 = 137.820804$ kN/mm in full fracture Job 1404933, $\Delta K_0 = -0.09\%$, $\Delta F_{\max} = -1.64\%$).
- Gate 5 – Native Remeshing Reproduction (71,320 vs $\approx 13,941$): SUPERVISOR_ACCEPTED_REPRODUCTION_LIMITATION_CLOSED.** The supervisor accepted that publication details do not expose enough information to reproduce every reported element count exactly. Parametric sensitivity trends are preserved and strictly disambiguated between offline preliminary sizing (1.0% $\rightarrow$ 71,320; 2.0% $\rightarrow$ 17,687; 3.0% $\rightarrow$ 8,120; 5.0% $\rightarrow$ 4,356) and the audited production batch meshes (1.0% $\rightarrow$ 71,320; 2.0% $\rightarrow$ 15,396; 3.0% $\rightarrow$ 7,633; 5.0% $\rightarrow$ 4,194 finite elements).
- Gate 6B – Mode-I Energetic and Multifaceted Convergence: ACTIVE – HIGHEST PRIORITY.** Decoupled evaluation across structural stiffness K_0 , peak force $F_{\max}$, peak displacement u_{peak} , trapezoidal boundary work W_{trap} , UEL energy terms (E_{elas} , E_{frac}), crack-path symmetry, temporal discretization (T_1 – T_3 and adaptive $2\times$), spatial discretization (S_1 – S_4), and length scale (l_0).
 - UEL Energy Output:** Formally promoted to UEL_ENERGY_OUTPUT_QUALIFIED_MECHANICALLY_NONINVASIVE via full 7,000-increment point-by-point mechanical parity audit ($\Delta K_0 = 0.000000\%$, $\Delta F_{\max} = 0.000000\%$).
 - Temporal Convergence:** Pre-peak response and peak characteristics are TEMPORALLY_STABLE ($\Delta K_0 = +0.000302\%$, $\Delta F_{\max} = -0.0229\%$, QUALIFIED_OVER_PREPEAK_INTERVAL_ONLY), while post-fracture wake exhibits increment-scale normalization sensitivity (TEMPORALLY_SENSITIVE_POSTPEAK).
 - Active Solver Solves on /scratch9/:** Actively tracking spatial candidate Job 1410179 (58k FE), convergence-control diagnostic Job 1410180 ($C_n = 0.50$), and errorTarget fracture batch Jobs 1410357–1410359 (ET2, ET3, ET5).
- Global Energy Identity: GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED.** The staggered solution scheme evaluates history $\mathcal{H}_n$ at step n and damage d_{n+1} at step $n+1$, disproving the existence of a joint discrete scalar potential ($\partial^2 \Pi / \partial \mathbf{u} \partial d \neq \partial^2 \Pi / \partial d \partial \mathbf{u}$).
- Active Scope Holds:** Mode-II, Gate 6C (State Transfer), Gate 7 (Visualization integration), and multi-rank MPI remain on **strict hold** pending supervisor acceptance of Gate 6B closure.

1.2 Key scientific findings presented in this package

- Energy Source Mechanical Parity:** Formally qualified as UEL_ENERGY_OUTPUT_QUALIFIED_MECHANICALLY_NONINVASIVE. Global mechanical force trajectories match bitwise ($|\Delta F| = 0.000000$ kN, $\Delta K_0 = 0.000000\%$) across all 7,000 increments (21,120 Newton iterations, 0 cutbacks) against uninstrumented baselines.
- Single-IP Deduplication Rule:** Companion visualizer UMAT elements replicate scalar whole-element energies into all 4 Gauss integration points. Summing across all IPs creates an artificial $4\times$ overcounting error. Enforcing

extraction at Integration Point 1 (IP1) recovers the once-per-underlying-finite-element global energy sum under the 2D implicit-unit-thickness convention.

- **Structural Stiffness Invariance:** Initial stiffness K_0 is exceptionally stable across spatial meshes (S_1 – S_4 : 137.95 to 137.84 kN/mm, spread $< 0.09\%$), native adaptive meshes (A_1 – A_4 : spread $< 0.09\%$), and time steps ($\Delta K_0 = +0.000302\%$), confirming kinematic, geometric, and elastic consistency.
- **Peak Force Mesh Sensitivity:** Peak reaction force drops monotonically with spatial mesh refinement from 0.7578 kN (S_1 , $h/l_0 = 0.400$) to 0.7312 kN (S_4 , $h/l_0 = 0.167$), extending to 0.7255 kN on the historical fine mesh ($h = 1.0 \mu\text{m}$, -4.26% total variation), classified as **MESH-SENSITIVE**.
- **Crack-Path Symmetry and Profile Convergence:** The damage-weighted crack-centroid trajectory deviates from the symmetry plane $y = 0.500 \text{ mm}$ by $\leq 3.10 \mu\text{m}$ across all fixed and adaptive meshes, confirming pure Mode-I symmetry within discrete element bounds.
- **Completed 3-Point Length-Scale Study (l_0):** Evaluated across $l_0 \in \{7.5, 11.25, 15.0\} \mu\text{m}$ on fixed S_3 mesh (Jobs 1406017.mm, 1406895.mm, 1406896.mm). K_0 is stable ($137.86 \rightarrow 137.68 \text{ kN/mm}$, 0.13% spread), while F_{max} ($0.7322 \rightarrow 0.6895 \text{ kN}$) and u_{peak} ($5.633 \rightarrow 5.579 \mu\text{m}$) are length-scale sensitive.

2. Mode-I Benchmark Definition and Canonical S1 Reference Baseline

2.1 Benchmark Boundary Value Problem

The Mode-I benchmark is defined on a two-dimensional square domain $\Omega = [0, 1.0 \text{ mm}] \times [0, 1.0 \text{ mm}]$ containing an initial sharp horizontal edge crack of length $a_0 = 0.50 \text{ mm}$ along the symmetry plane $y = 0.50 \text{ mm}$ (Fig. 1). The crack is discretized as a zero-gap seam with duplicate coincident nodal lines, preserving exact geometric compliance.

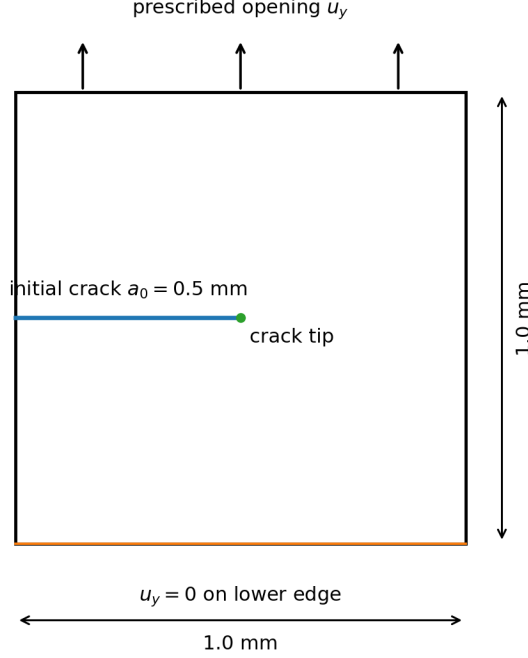


Figure 1: Mode-I benchmark boundary value problem: 1.0 mm $\times$ 1.0 mm square domain with a sharp edge crack of length $a_0 = 0.5 \text{ mm}$ along $y = 0.5 \text{ mm}$. Boundary conditions enforce roller support ($u_y = 0$) on the bottom edge with a pinned point ($u_x = 0$) to eliminate rigid-body motion, and uniform tensile displacement $u_y = u$ on the top edge.

Scientific conclusion. This figure fixes the governed physical boundary-value problem. All later fixed and adaptive simulations use the same geometry, supports, and loading, so their differences can be attributed to discretization, time stepping, or adaptivity rather than to a changed model definition.

The material and phase-field constitutive parameters follow Pandey and Kumar [2025]:

- Young’s modulus: $E = 210.0 \text{ GPa} = 210.0 \text{ kN/mm}^2$;
- Poisson’s ratio: $\nu = 0.30$;
- Critical fracture energy release rate: $G_c = 2.7 \times 10^{-3} \text{ kN/mm}$ (2.7 N/mm);
- Phase-field regularization length scale: $l_0 = 0.0075 \text{ mm} = 7.5 \mu\text{m}$;
- Residual stiffness parameter: $k_{\text{res}} = 1.0 \times 10^{-7}$.

2.2 Boundary-Condition Verification and Seam Compliance

A vital early physical distinction is that global initial structural stiffness $K_0 = \Delta F / \Delta u$ is a structural property, distinct from the material Young’s modulus E . During initial model verification, representing the crack as a finite-width notch was found to introduce substantial artificial compliance ($K_0 \approx 75.47 \text{ kN/mm}$). The geometry was therefore corrected to the required zero-gap sharp seam before the governed convergence and adaptive-remeshing studies were performed. All subsequent results use the corrected sharp-seam geometry.

2.3 Canonical S1 Baseline Provenance

The governed canonical baseline for the Mode-I benchmark is Job 1406015.mm_{master02} (S_1 , 15,192 underlying finite elements, 15,522 total *NODE entries (15,521 mesh nodes + RP 999999)), generated from the canonical input deck (SHA256: bc000e0c5791150a711cb7af447a0bb34d1b9d9af5c6ce9d096c98749e8edf61). Extracted from the audited source file S1_h0030_15k_MECHANICAL_FU_AUDITED.csv (SHA256: E3100B7429E14B8BCEAB9952DDF8B0419CA9F71DF058F5916668922185AA25E2) over the initial $N = 400$ active increments ($u \leq 1.0 \mu\text{m}$), unconstrained ordinary least squares yields:

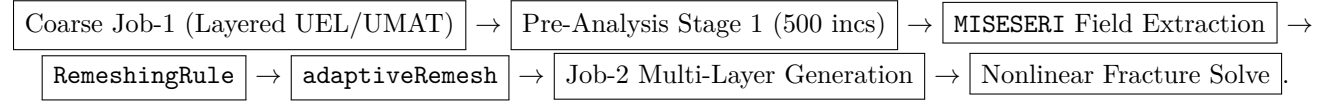
$$K_0 = 137.945520 \text{ kN/mm}, \quad b = 4.472368 \times 10^{-5} \text{ kN}, \quad R^2 = 0.99999960. \quad (1)$$

The peak reaction force is $F_{\text{max}} = 0.757778 \text{ kN}$ at displacement $u(F_{\text{max}}) = 0.005857 \text{ mm}$, matching digitized literature values from Pandey and Kumar [2025] ($F_{\text{max}} \approx 0.758 \text{ kN}$, $u \approx 0.005860 \text{ mm}$). Reconstructed Job 1398090.mm_{master02} is preserved as historical baseline evidence. This canonical response provides the quantitative anchor against which all spatial, temporal, and adaptive remeshing results are judged.

3. Pandey–Kumar Adaptive Refinement and MISESeri Mechanism

3.1 Two-Job Adaptive Refinement Procedure

Pandey and Kumar [2025] employ an automated two-job pre-refinement procedure:



Abaqus-native error-guided pre-refinement workflow

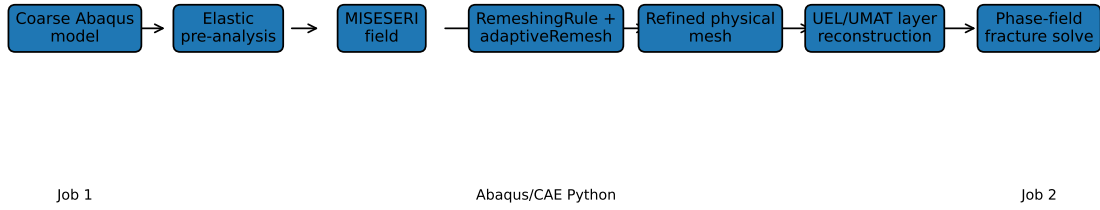


Figure 2: Pandey–Kumar two-job adaptive pre-refinement workflow: a coarse pre-analysis on the layered UEL/UMAT model evaluates stress recovery error on the co-located continuum layer, which drives native `adaptiveRemesh` before reconstructing the multi-layer UEL/UMAT phase-field model.

Scientific conclusion. The method is an error-driven *pre-refinement* workflow, not remeshing during crack propagation: the mesh is generated before the nonlinear phase-field fracture solve begins.

The published Mode-I description outlines a 500-increment first loading stage followed by a 1000-increment second stage. However, the exact `stepName` and frame selection semantics used within the native `RemeshingRule` are not explicitly disclosed in the publication.

3.2 Physical Understanding and Provenance of MISESeri

MISESeri is the Abaqus stress discretization error indicator associated with the recovered von Mises stress field in linear continuum elasticity [Abaqus, 2023]. It is based on stress-recovery error estimation on the linear-elastic continuum stress field; it is **not** phase-field error, **not** damage error, and **not** fracture energy imbalance.

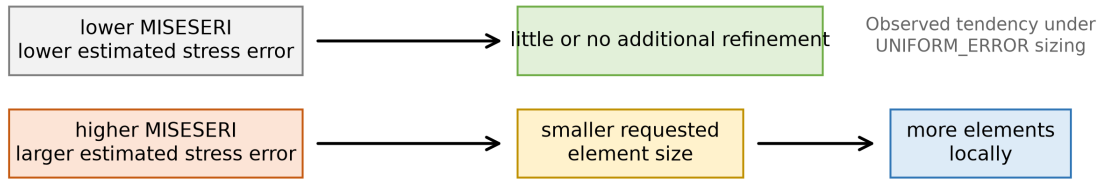


Figure 3: Directional relationship of MISESeri under uniform-error sizing: elevated stress error near singularities dictates local refinement without representing a direct algebraic formula to element size.

Scientific conclusion. Higher recovered-stress discretization error promotes stronger local refinement, but `errorTarget` is a target tolerance rather than a direct MISESeri threshold or an explicit formula for element size.

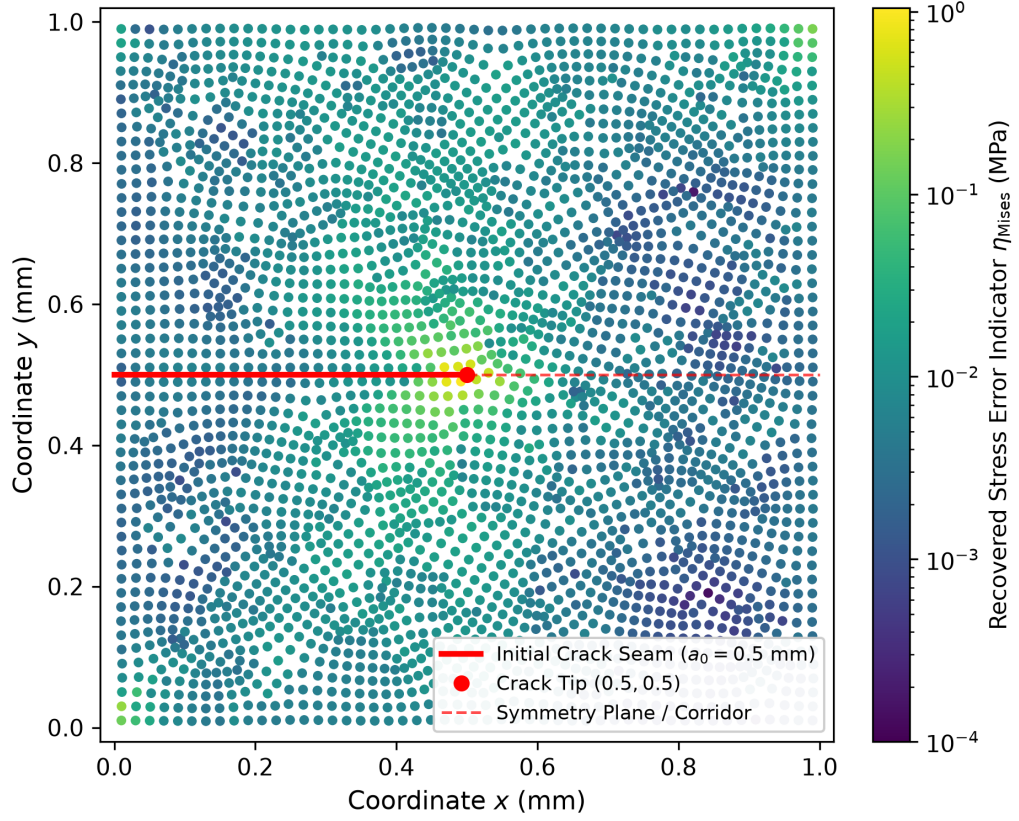
The sizing parameter `errorTarget=1.0` in Listing 1 of Pandey and Kumar [2025] denotes a 1.0% target error tolerance under `UNIFORM_ERROR` sizing. It does **not** mean `MISESeri = 1.0` or that 1% of elements are refined. Similarly, `errorTarget=2.0` specifies a 2.0% target tolerance, not `MISESeri = 2.0`.

3.3 Canonical Mode-I Pre-Analysis and Spatial Association

The canonical Mode-I pre-analysis contains 2,906 finite elements (2,818 CPE4 + 88 CPE3), 2,988 nodes, and exactly 2,906 scalar `WHOLE_ELEMENT` MISESeri indicator values (Fig. 4).

Applying `errorTarget=1.0%` with `coarseningFactor=NOT_ALLOWED` generates the 71,320-element mesh (Fig. 5). As shown in Fig. 6, adapted element sizes correlate closely with the coarse indicator field.

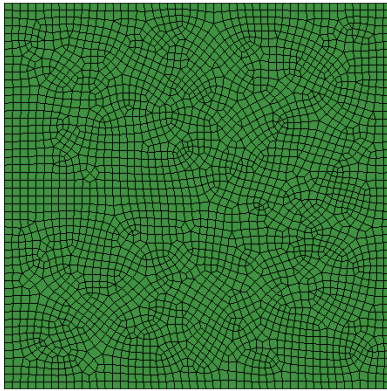
Figure 3.1: Canonical Mode-I Coarse Mesh MisesERI Distribution
 $(N_{\text{elem}} = 2,906, 5,812 \text{ Duplicate-Collapsed Values, Abaqus 2023})$



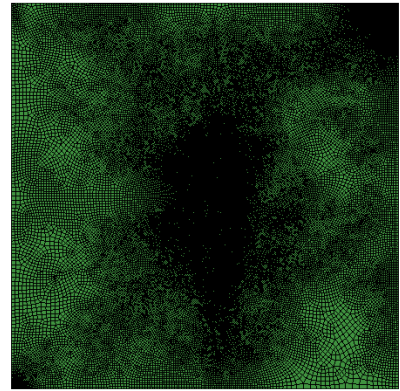
Note: MisesERI is a recovered von Mises stress discretization error indicator (Zienkiewicz-Zhu stress recovery); it is NEITHER phase-field error NOR damage error.

Figure 4: Canonical Mode-I MisesERI error indicator field evaluated on the 2,906-element coarse mesh, showing error concentration at the initial crack tip.

Scientific conclusion. The coarse elastic mesh is most under-resolved around the initial crack tip, where the stress gradient is highest. MisesERI is a stress discretization-error indicator, not a phase-field or damage-error measure.



(a) Coarse pre-analysis mesh (2,906 elements)



(b) Adapted mesh (71,320 elements)

Figure 5: Visual comparison of (a) initial coarse mesh (2,906 elements) and (b) native adapted mesh (71,320 elements).

Scientific conclusion. Abaqus converts the nonuniform error field into the expected nonuniform mesh: element density is concentrated along the crack-tip corridor while regions farther away remain comparatively coarse.

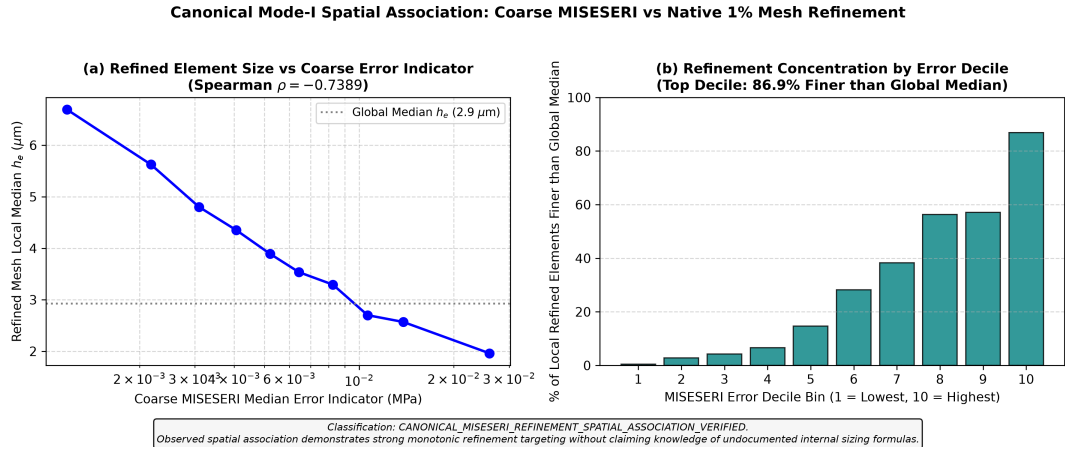


Figure 6: Empirical spatial association between coarse MISESeri and adapted element size h for the Mode-I benchmark.

Scientific conclusion. Large coarse-mesh MISESeri values are spatially associated with small adapted elements. This verifies error-guided refinement without claiming an exact proprietary MISESeri $\rightarrow h$ mapping law.

4. Resolution of the 71,320-Element Boundary Formatting Defect

4.1 Defect Mechanism and Verified Resolution

An early 71,320-element adaptive model (historical Job 1399632) exhibited an artificial stiffness deficit caused by an Abaqus keyword input card limit: the bottom-edge node set `N_BOTTOM` contained 150 node identifiers written on a single data line without `GENERATE`. Abaqus `pre` parsed only the first 16 node entries and silently omitted the remaining 134 nodes from the constraint set.

Wrapping the `N_BOTTOM` data across valid data lines (≤ 16 entries per line) fully constrained all 150 intended bottom nodes, restoring the structural stiffness to the reference range. The defect was independently verified and is formally classified as **RESOLVED_AND_CLOSED**; all subsequent adaptive results use the corrected boundary-condition definition.

The two authoritative verification runs confirm full mechanical recovery (Table 1 and Fig. 7):

- **Corrected Full Fracture (Job 1404933.mm.master02):** Solved the complete fracture process, achieving $K_0 = 137.820804 \text{ kN/mm}$ ($\Delta K_0 = -0.09\%$) and peak load $F_{\max} = 0.745325 \text{ kN}$ ($\Delta F_{\max} = -1.64\%$).
- **Frozen Elastic Requalification (Job 1405044.mm.master02):** Independent elastic verification achieving $K_0 = 138.021013 \text{ kN/mm}$ ($\Delta K_0 = +0.05\%$) with zero node lift and zero warnings.

Table 1: Mechanical response recovery after resolving the `N_BOTTOM` keyword line limit.

Case Description	Job ID	Elements	K_0 [kN/mm]	$F_{\max}$ [kN]	Status / Governed Role
Fixed Reference Anchor (S_1)	1406015	15,192	137.945520	0.757778	Canonical reference anchor
Corrected Full Fracture (A_1)	1404933	71,320	137.820804	0.745325	Production 1% adaptive mesh
Frozen Elastic Requalification	1405044	71,320	138.021013	–	Independent elastic verification

Force--Displacement Comparison: Canonical S_1 Reference vs Corrected A_1 Adaptive Mesh

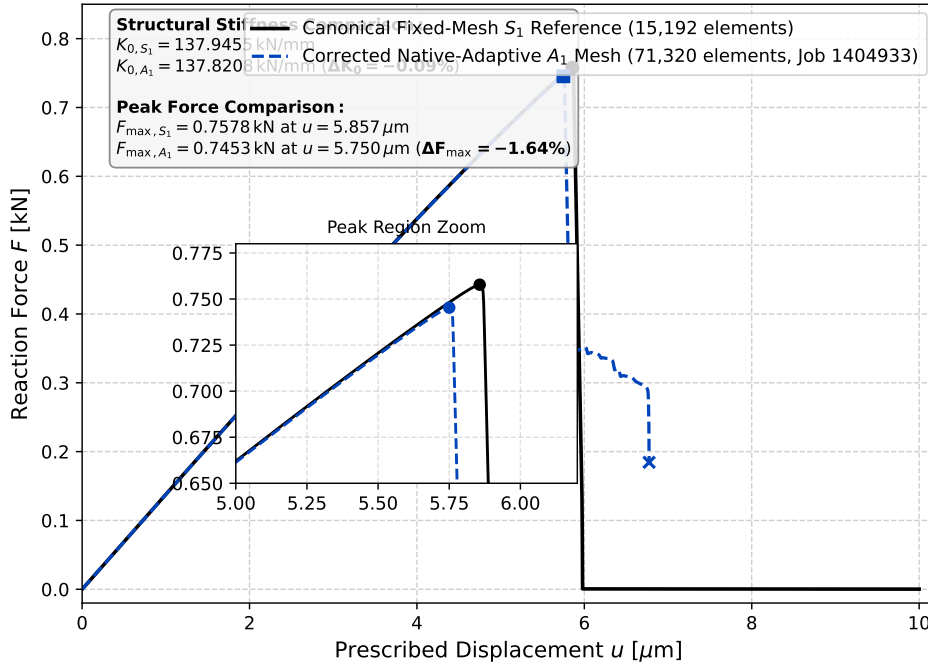


Figure 7: Force–displacement comparison between the canonical fixed-mesh S_1 reference (15,192 finite elements) and the corrected native-adaptive A_1 model with `errorTarget = 1%` (71,320 finite elements). The initial structural stiffness differs by only 0.09%, while the adaptive peak reaction force is approximately 1.64% below the reference value.

Scientific conclusion. Correcting `N_BOTTOM` restores the adaptive model’s elastic stiffness. The remaining peak-force difference is therefore a fracture/discretization effect, not evidence of an unresolved elastic boundary-condition defect.

5. Discrepancy Audit: 71,320 vs 13,941 Elements and Governing Decision

5.1 Publication Discrepancy and Comprehensive Investigation

Pandey and Kumar [2025] report an adapted Mode-I element count of $\approx 13,941$, while strictly applying the published Listing 1 settings to the full domain generates 71,320 finite elements.

An exhaustive numerical and software investigation established:

1. **Abaqus Release Invariance:** Abaqus 2019 GA, 2021.HF26, 2022 GA, and 2023.HF4 all generate 100.000% bitwise identical 71,320-element meshes with matching SHA-256 hashes.
2. **Single-Factor Parameter Testing:** Testing load magnitude, output frequency, boundary conditions, and mesher controls ruled out trivial parameter differences.
3. **Regional Breakdown:** The crack process corridor ($|y| \leq 0.05$ mm) contains 9,061 elements (12.7%), while far-field and transition regions contain 62,259 elements (87.3%) because coarsening was disabled (`coarseningFactor=NOT_ALLOWED`).
4. **Pre-Analysis Loading and Step Lineage:** Evaluating native remeshing against the qualified two-step layered UEL/UMAT pre-analysis (Job 1409554.mm`master02`) deterministically yields 48,329 finite elements (48,093 nodes) under 1.0% error target (100.000% mesh identity across independent repeats). An interim provisional run (Job 1409546.mm`master02`) yielded 42,318 elements.

5.2 Supervisor Decision and Definitive Closure

During the meeting on 17 September 2026, the supervisor formally closed this investigation under the following governing ruling:

SUPERVISOR_ACCEPTED_REPRODUCTION_LIMITATION_CLOSED

The supervisor confirmed that:

- The complete author code, internal scripts, and exact frame selection routines are unavailable;
- The publication does not disclose sufficient information to reproduce the specific 13,941 element count;
- The reproducible 48,329-element mesh is classified as **PROJECT_REPRODUCIBLE_STEP1_1PCT_RESULT**, not an authoritative literature match;
- The remaining difference between 48,329 and 13,941, along with the exact step/frame filtering semantics, is formally designated as **UNRESOLVED_PUBLICATION_IMPLEMENTATION_DETAIL** and **FRAME_SELECTION_SEMANTICS_UNRESOLVED**;
- Exact 13,941 reproduction is **not an active blocker**;
- Reproduction of general error-guided adaptivity trends is fully sufficient;
- Additional effort aimed solely at discovering an unpublished detail is not justified;
- All tested sensitivity results shall be preserved in the thesis as documented evidence.

5.3 Preserved Parametric Sensitivity Trends

Table 2 summarizes the preserved sensitivity results across four target error tolerances, demonstrating monotonic mesh reduction as tolerance relaxes.

Table 2: Parametric sensitivity of adapted finite element counts to the `errorTarget` tolerance.

Case	<code>errorTarget</code>	Coarse Baseline Count	Corrected Pre-Analysis Count	Classification
A_1	1.0%	71,320	48,329	Reproducible Step-1 1% result
A_2	2.0%	17,687	11,737	2.0% error tolerance
A_3	3.0%	8,120	5,158	3.0% error tolerance
A_4	5.0%	4,356	3,763	5.0% error tolerance

5.4 Influence of `errorTarget` on the Final Stage-14 Step-2 Adaptive Refinement

To systematically establish the spatial localization and element-count sensitivity of the native Abaqus `adaptiveRemesh` engine when evaluated on the true localized phase-field damage state, a rigorous parametric sweep across `errorTarget` $\in \{1.0\%, 2.0\%, 3.0\%, 5.0\%\}$ was conducted targeting Step-2 ($u_y = 0.010$ mm, Frame 1022) of the companion pre-analysis ODB (PK_M1_JOB1_INF_COMPANION_2906.odb). All sizing parameters were kept identical ($h_{\min} = 1.0 \mu\text{m}$, $h_{\max} = 20.0 \mu\text{m}$, `refinementFactor` = 10, `coarseningFactor` = NOT_ALLOWED, `region` = ALL_ELEM).

Table 3 presents the quantitative metrics across the four evaluated error targets.

**Stage-14 Step-2 Native Adaptive Remeshing Sensitivity: errorTarget in {1.0%, 2.0%, 3.0%, 5.0%}
Evaluated on Localized Phase-Field Companion Pre-Analysis ODB (Step-2, Frame 1022)**

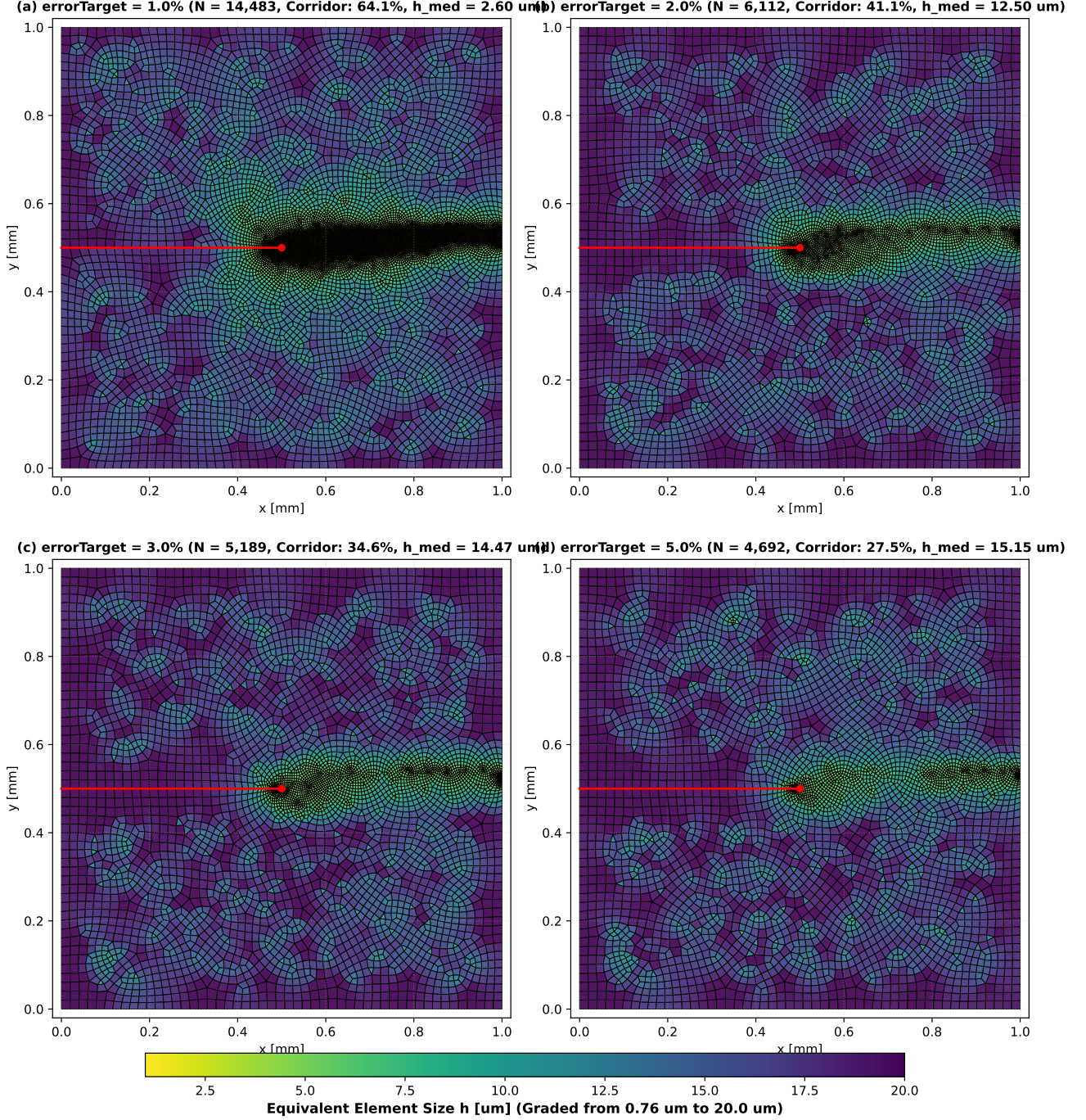


Figure 8: Parametric sensitivity of native Abaqus `adaptiveRemesh` across four error targets ($errorTarget \in \{1.0\%, 2.0\%, 3.0\%, 5.0\%\}$) evaluated on the localized Stage-14 Step-2 phase-field pre-analysis ODB. Color contours depict equivalent element size $h_{eq} \in [0.76, 20.0] \mu m$. ET1 retains dense, target-like crack-ligament refinement (64.12% corridor share), whereas ET2, ET3, and ET5 progressively under-refine the horizontal propagation path.

Table 4: Stage-14 Step-2 errorTarget fracture batch execution status and predeclared comparison matrix.

Case Identifier	ET1 (1.0%)	ET2 (2.0%)	ET3 (3.0%)	ET5 (5.0%)
Base Elements (N_{base})	14,483	6,112	5,189	4,692
Total 3-Layer Elements (N_{tot})	43,449	18,336	15,567	14,076
Total Nodes (N_{nodes})	14,456	6,181	5,262	4,759
PBS Job ID	1409982.mmaste02	1410357.mmaste02	1410358.mmaste02	1410359.mmaste02
Execution Mode	Serial 1-CPU	Serial 1-CPU	Serial 1-CPU	Serial 1-CPU
Scratch Path	/scratch9/...	/scratch9/...	/scratch9/...	/scratch9/...
Datacheck Result	Exit 0 (Passed)	Exit 0 (Passed)	Exit 0 (Passed)	Exit 0 (Passed)
Native Mesh Verdict	TARGET_LIKE	AWAY_FROM_TARGET	AWAY_FROM_TARGET	AWAY_FROM_TARGET
Corridor Share ($ y - 0.5 \leq 0.05$ mm)	64.12%	41.07%	34.59%	27.51%
Initial Stiffness K_0 [kN/mm]	137.909558 (−0.026%)	Evaluating...	Evaluating...	Evaluating...
Peak Reaction Force F_{max} [kN]	0.743701 (−1.858%)	Evaluating...	Evaluating...	Evaluating...
Peak Displacement $u(F_{\text{max}})$ [mm]	0.005733	Evaluating...	Evaluating...	Evaluating...
Terminal Displacement u_{term} [mm]	0.007889	Evaluating...	Evaluating...	Evaluating...
Fracture Functional E_{frac} [mJ]	2.285469	Evaluating...	Evaluating...	Evaluating...
Bookkeeping Residual $\varepsilon_{\text{book}}$	1.1048%	Evaluating...	Evaluating...	Evaluating...
Fracture Response Status	RESPONSE_STABLE	RUNNING / PENDING	RUNNING / PENDING	RUNNING / PENDING

6. Multifaceted Mode-I Convergence Evaluation

In accordance with supervisor directives, mesh adequacy and numerical convergence are evaluated across multiple complementary physical, spatial, temporal, and energetic quantities rather than peak reaction force alone.

6.1 Spatial Mesh Resolution Series (S_1 – S_4)

Four uniform meshes spanning element edge lengths from $h = 3.0\ \mu\text{m}$ (S_1 , 15,192 elements) down to $h = 1.25\ \mu\text{m}$ (S_4 , 51,408 elements) evaluate spatial discretization characteristics ($h/l_0 \in [0.167, 0.400]$ at fixed $l_0 = 7.5\ \mu\text{m}$), alongside a separate historical fine-mesh result ($h = 1.0\ \mu\text{m}$, 69,384 elements, Job 1406019.mmaster02):

1. **Initial Structural Stiffness K_0 is Stable:** Across the governed S_1 – S_4 spatial series, K_0 varies from 137.945520 to 137.841200 kN/mm. An additional historical $h = 1.0\ \mu\text{m}$ mesh gives 137.823301 kN/mm, confirming that the initial structural stiffness remains exceptionally stable over the broader tested resolution range ($< 0.09\%$ total variation, $R^2 > 0.999999$). Classified as **CONVERGED / STABLE**.
2. **Peak Force F_{max} and Displacement u_{peak} are Mesh-Sensitive:** Across the full tested range, peak force decreases monotonically from 0.7578 kN (S_1) to 0.7255 kN (historical fine mesh), a total reduction of 4.26% (Fig. 9 and 10). The displacement at peak load shifts from 5.857 to 5.575 μm , corresponding to an approximately 4.81% earlier peak with refinement. Classified as **MESH-SENSITIVE**.

Mode-I Tensile Force-Displacement Response (Spatial Series S_1 – S_4 + Historical Fine Mesh)

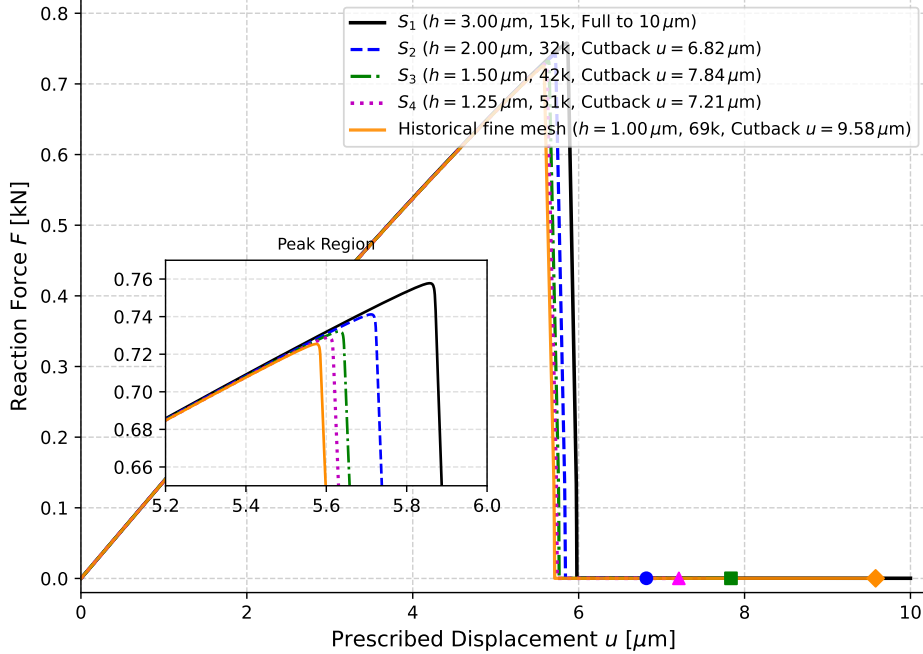


Figure 9: Mode-I tensile force-displacement response across uniform mesh resolutions (S_1 – S_4 plus historical fine mesh), showing identical elastic stiffness and systematic peak load mesh sensitivity.

Scientific conclusion. The elastic slopes are effectively mesh-independent, but refinement systematically lowers F_{max} and shifts u_{peak} earlier. Elastic response is stable; fracture initiation remains spatially mesh-sensitive.

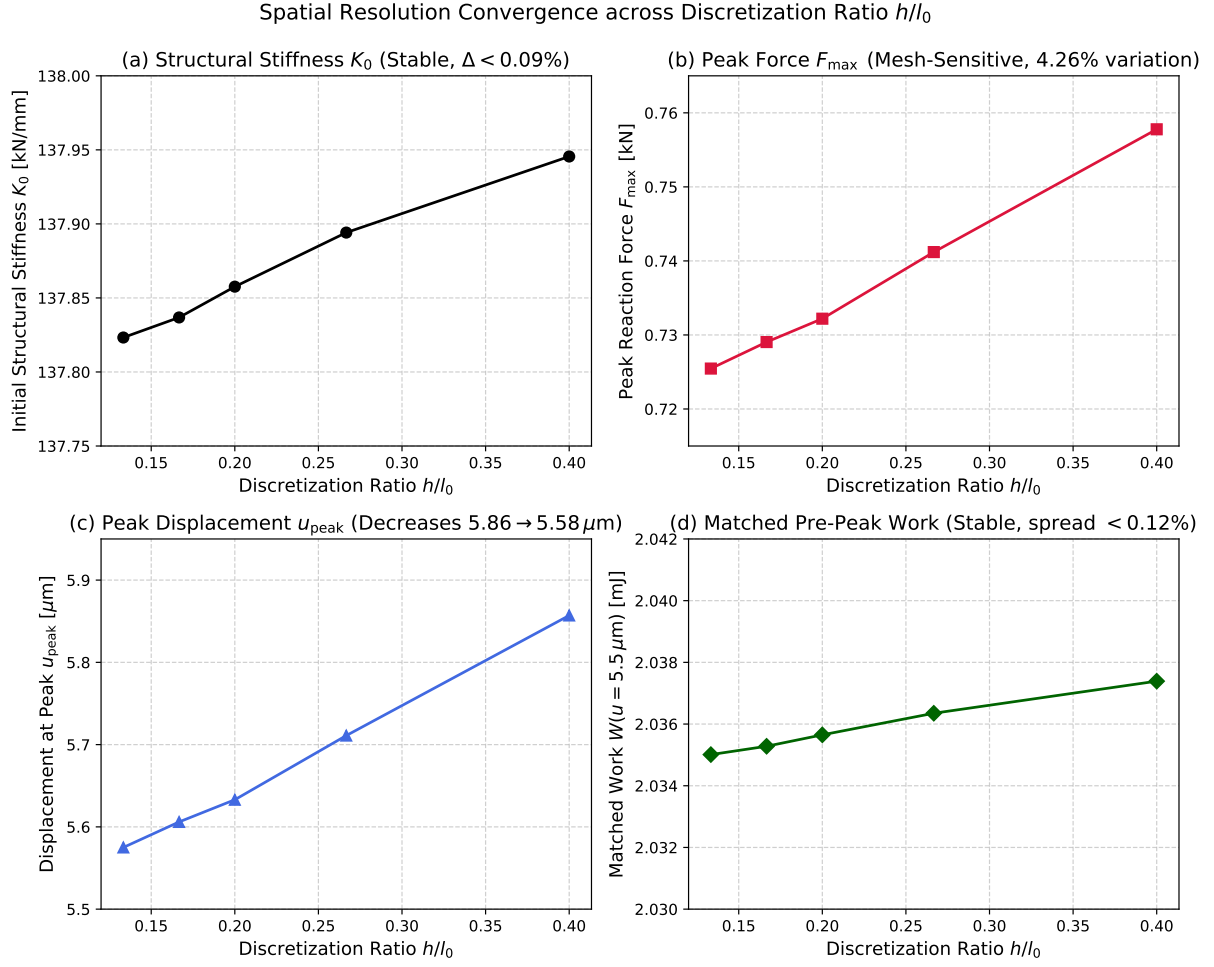


Figure 10: Discretization ratio (h/l_0) scaling metrics: (a) exceptional invariance of initial stiffness K_0 ($\Delta < 0.09\%$); (b) monotonic mesh sensitivity of peak force $F_{\max}$ (4.26%); (c) peak displacement shift u_{peak} ; (d) matched-displacement external work W_{trap} .

Scientific conclusion. Convergence is quantity-dependent: K_0 and matched pre-peak work are nearly invariant, whereas $F_{\max}$ and u_{peak} change systematically with h/l_0 . The solution must not be described as fully mesh-converged.

6.2 Temporal Time-Increment Scaling (T_1 – T_3)

Time incrementation was tested across three schedules at fixed baseline resolution ($h = 3.0 \mu\text{m}$): Case T_1 (coarse $2\times$, 3,500 incs), Case T_2 (nominal $1\times$, 7,000 incs), and Case T_3 (fine $0.5\times$, Job 1406317.mm_{master02}, completed all 10,022 Step-2 incs to $u = 10.0 \mu\text{m}$).

- **Peak Mechanical Invariance:** Initial stiffness K_0 varies by $< 0.001\%$, and peak force varies by $< 0.07\%$ ($0.75815 \rightarrow 0.75763 \text{ kN}$), proving temporal stability of limit loads (**CONVERGED / STABLE**, Fig. 11).
- **Energetic Temporal Sensitivity:** Terminal trapezoidal boundary work W_{trap} varies by 3.24% ($2.41011 \rightarrow 2.33189 \text{ mJ}$), and the two-term bookkeeping difference $\mathcal{R}_{\text{bookkeeping}} = W_{\text{trap}} - (E_{\text{elas}} + E_{\text{frac}})$ shifts from $+0.378\%$ to $+3.541\%$ (**TEMPORALLY SENSITIVE**).

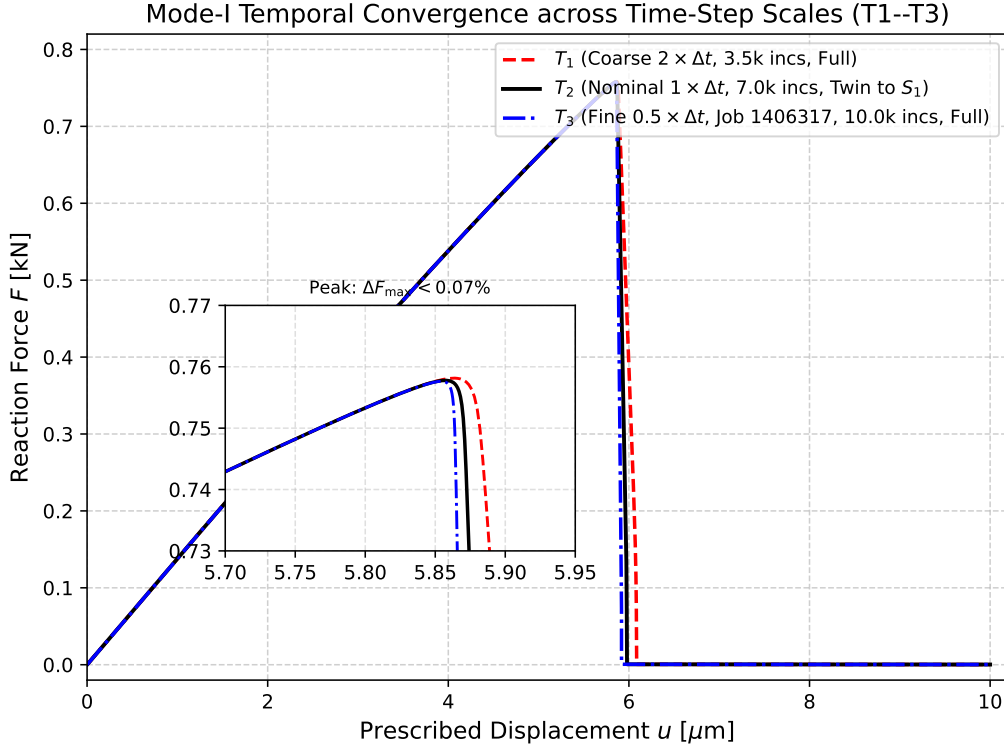


Figure 11: Mode-I temporal convergence across time increment scales (T_1 : $2\times$, T_2 : $1\times$, T_3 : $0.5\times$, Job 1406317.mm_{master02}), showing step-size invariance of the mechanical peak.

Scientific conclusion. The mechanical response through the peak is temporally converged over T_1 – T_3 ($\Delta K_0 < 0.001\%$, $\Delta F_{\text{max}} < 0.07\%$). The spatial peak sensitivity in Fig. 9 therefore cannot reasonably be attributed to insufficient time resolution; post-peak energy is assessed separately.

6.3 Adaptive Mesh Temporal Refinement Audit (2× Refinement: Job 1410027 vs 1409982)

To evaluate time-step convergence on the production 14,483-element adaptive mesh, a controlled 2× temporal refinement diagnostic (Job 1410027.mmaste02, 8,958 completed increments, Package 26) was audited against the baseline canonical ET1 solve (Job 1409982.mmaste02, 4,890 increments, Package 25). All physical parameters, 3-layer UEL subroutine source (CE8D5EDC...), and solver controls ($I_A = 10$, $\Delta t_{\min} = 10^{-9}$ s) were kept strictly invariant.

1. Elastic and Pre-Peak Temporal Stability:

- **Canonical Stiffness K_0 ($u \leq 1.0 \mu\text{m}$):** Baseline $K_0 = 137.909558 \text{ kN/mm}$ ($N = 400$, $\Delta u = 2.50 \text{ nm}$) vs 2× Refined $K_0 = 137.909975 \text{ kN/mm}$ ($N = 800$, $\Delta u = 1.25 \text{ nm}$), yielding $\Delta K_0 = +0.000302\%$ ($R^2 = 0.99999960$). Classified as **TEMPORALLY_STABLE**.
- **Peak Reaction Force $F_{\max}$ ($u \approx 5.73 \mu\text{m}$):** Baseline $F_{\max} = 0.743701 \text{ kN}$ ($u_{\text{peak}} = 5.733 \mu\text{m}$) vs 2× Refined $F_{\max} = 0.743530 \text{ kN}$ ($u_{\text{peak}} = 5.730 \mu\text{m}$), yielding $\Delta F_{\max} = -0.0229\%$ and $\Delta u_{\text{peak}} = -0.052\%$. Classified as **TEMPORALLY_STABLE**.
- **Pre-Peak Matched Parity:** Pointwise force discrepancy across all reached pre-peak states ($u \in [1.0, 5.733] \mu\text{m}$) is bounded by $|\Delta F|/F \leq 0.0436\%$ (Fig. 12 and 13).

2. **Energy Partitioning Dynamics at Peak:** Total model energy $E_{\text{model}} = E_{\text{elas}} + E_{\text{frac}}$ at peak displacement ($u = 5.730 \mu\text{m}$) is 2.205225 mJ in both models ($\Delta E_{\text{model}} = 0.000000 \text{ mJ}$ to 6 decimal places, Fig. 14).

3. **Common Displacement Interval & Zero Forward-Filling:** The common actually reached displacement domain is $u \in [0.0, 7.46970 \mu\text{m}]$. Unreached states ($u > 7.46970 \mu\text{m}$) are strictly classified as **NOT_REACHED** with zero forward-filling.

4. **Post-Fracture Severed-Wake Solver Stagnation Mechanism:** Both models experienced cut-back ceiling exhaustion ($I_A = 10$) down to $\Delta t_{\min} = 10^{-9}$ s in the severed wake ($x \approx 0.56 \text{ mm}$, $d \approx 0.999$, $\sigma \approx 0$). In the refined model, halving the nominal increment scale ($\Delta u_{\text{nom}} = 0.50 \text{ nm}$ vs 1.00 nm) halved Δu_{inc} , making the dimensionless displacement correction check $c_{\text{max}}/\Delta u_{\text{inc}}$ twice as strict, triggering cutback exhaustion at $u = 7.470 \mu\text{m}$ instead of $7.889 \mu\text{m}$ (Fig. 15). Classified as **POST_FRACTURE_CONVERGENCE_NORMALIZATION_SENSITIVITY_VERIFIED** and **TEMPORALLY_SENSITIVE_POSTPEAK**.

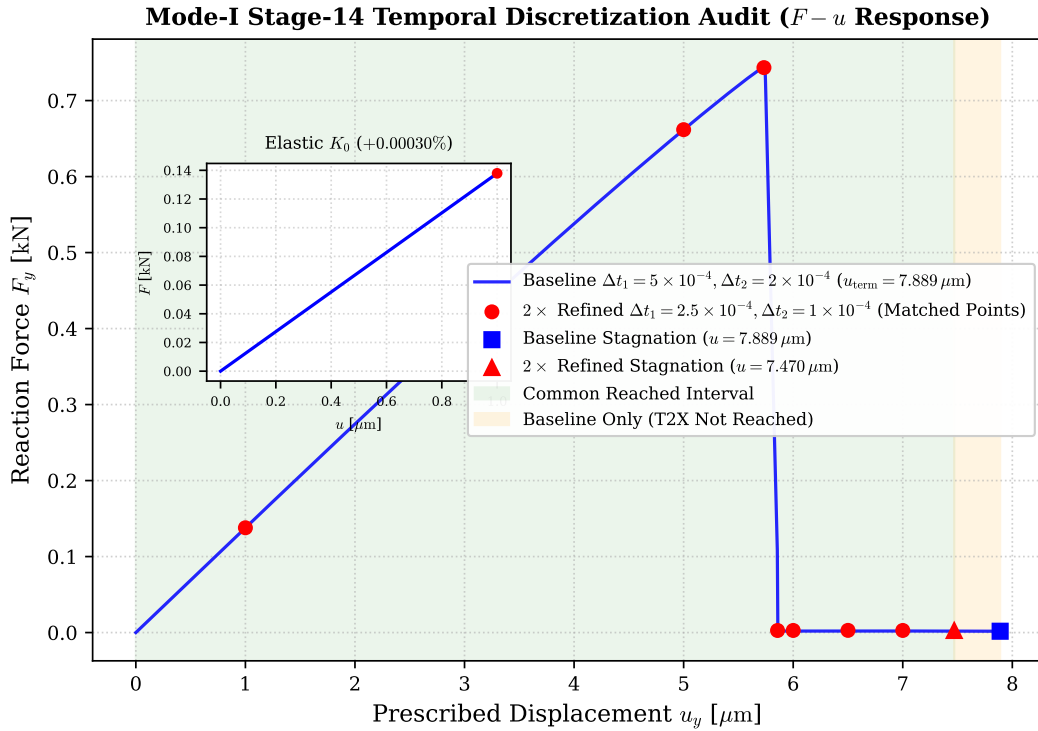


Figure 12: Mode-I adaptive mesh 2× temporal refinement audit: $F-u$ response comparing Baseline (4,890 incs, $u_{\text{term}} = 7.889 \mu\text{m}$) against 2× Refined (8,958 incs, $u_{\text{term}} = 7.470 \mu\text{m}$), highlighting common reached interval and pre-peak K_0 inset.

Scientific conclusion. Pre-peak mechanical response and peak load are fully temporally converged ($\Delta K_0 = +0.00030\%$, $\Delta F_{\max} = -0.0229\%$). The earlier termination at $u = 7.470 \mu\text{m}$ is a post-fracture wake normalization artifact, not physical instability.

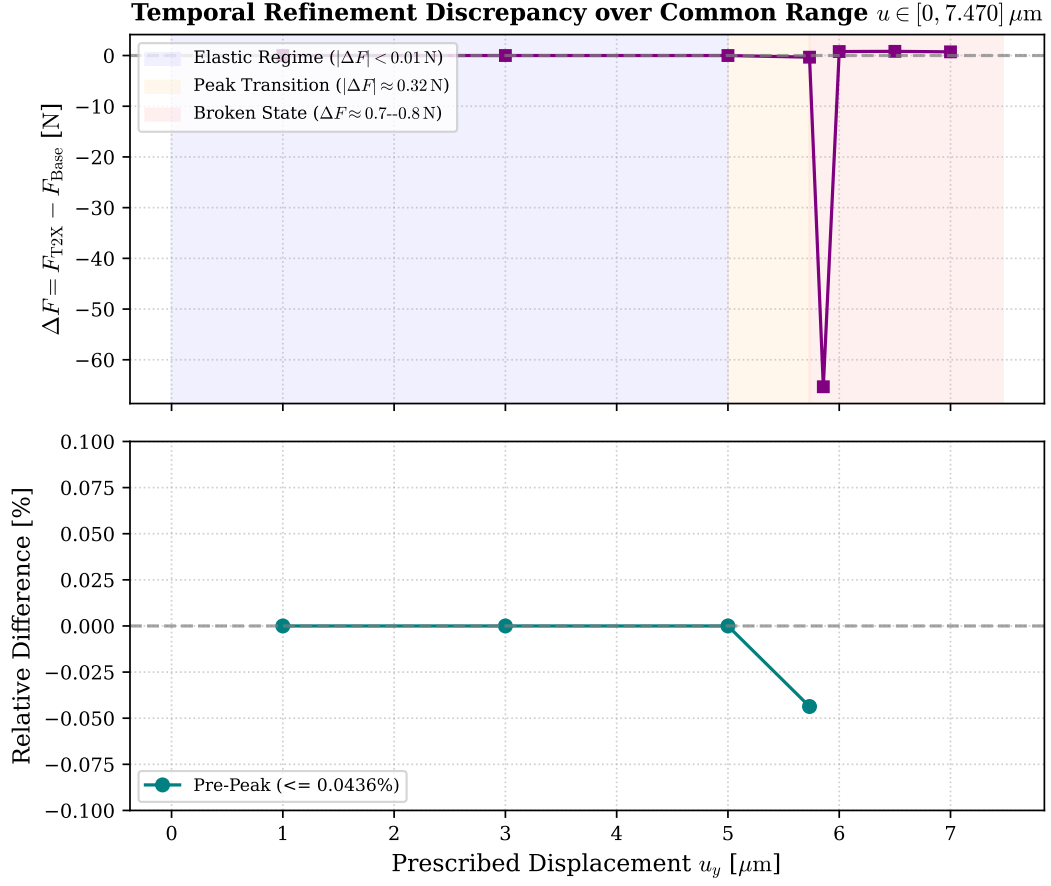


Figure 13: Pointwise force discrepancy ΔF and relative difference [%] over the common reached displacement interval $u \in [0, 7.470] \mu\text{m}$, separating elastic ($\leq 0.0436\%$) from post-peak broken-state regimes.

Scientific conclusion. Discrepancy is negligible in the elastic regime ($< 0.01 \text{ N}$), tight across peak load (0.32 N), and settles onto a minute post-fracture wake offset ($0.7\text{--}0.8 \text{ N}$) in the fully severed ligament.

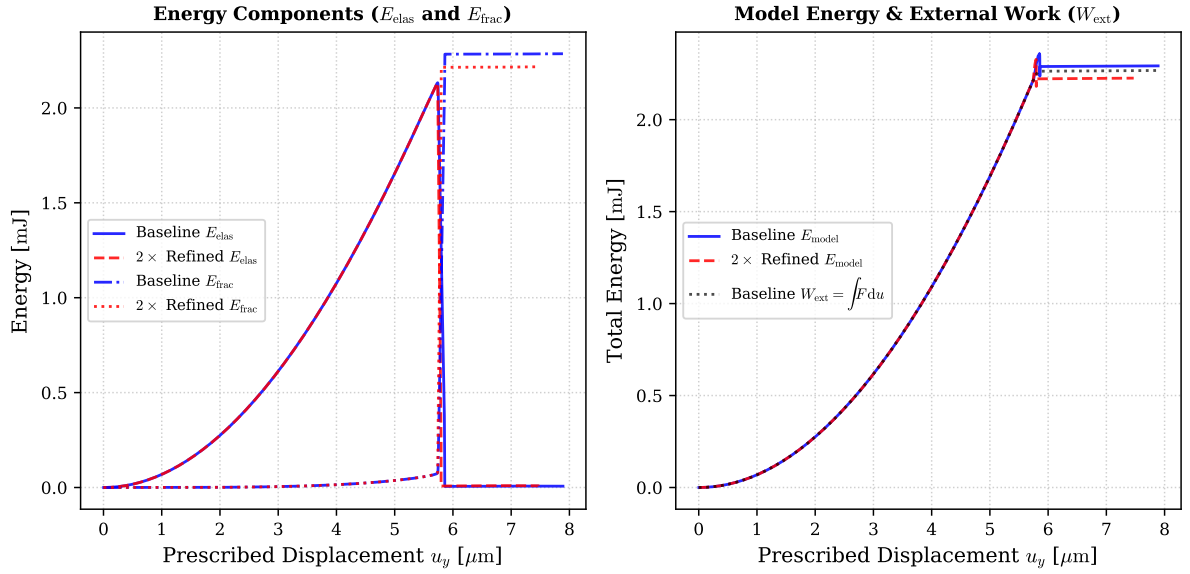


Figure 14: Matched energy evolution for baseline and $2 \times$ temporal refined models: (a) stored elastic energy E_{elas} and fracture functional E_{frac} ; (b) total model energy E_{model} and external work W_{ext} .

Scientific conclusion. Energy partitioning is bitwise identical through elastic loading and matches to 0.000000 mJ at peak load (2.205225 mJ). Broken-state fracture dissipation converges within 2.98% .

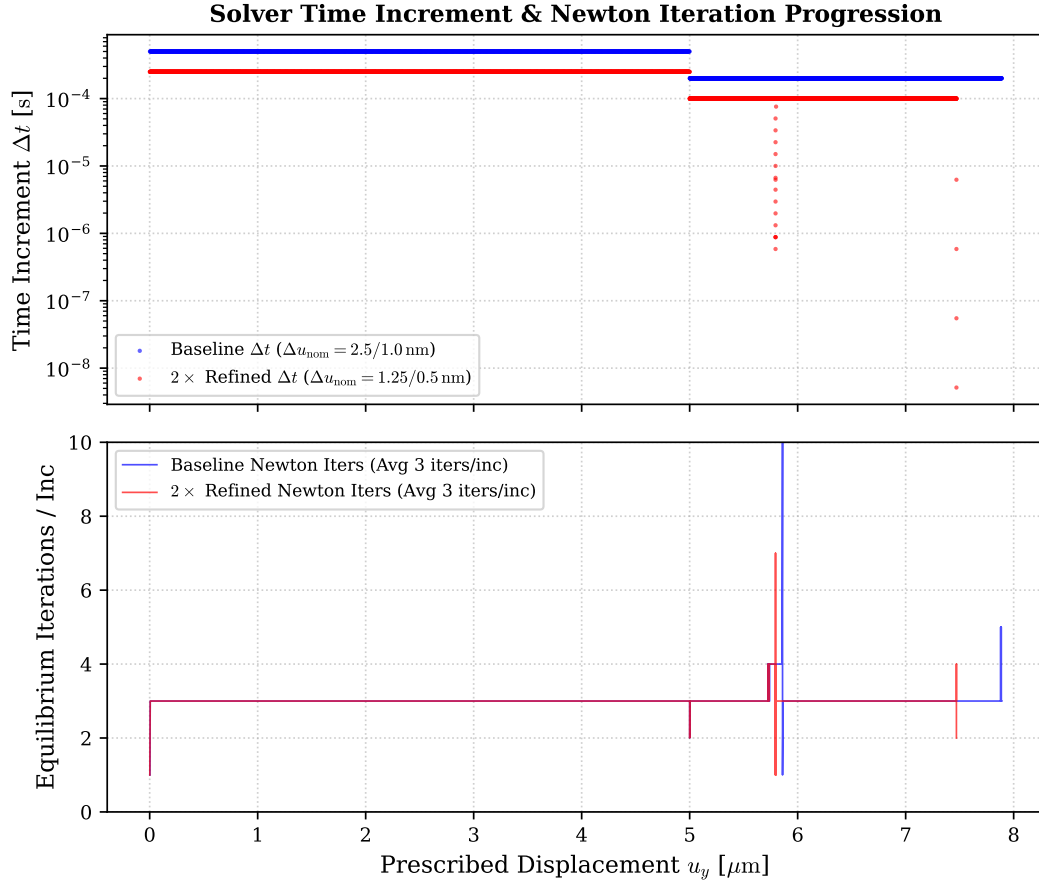


Figure 15: Solver time increment Δt progression (log scale) and Newton equilibrium iterations per increment across baseline and $2 \times$ refined runs, showing consistent 3 iters/inc across all reached states.

Scientific conclusion. Both simulations advance with steady 3 Newton iterations per increment throughout linear and softening phases, demonstrating zero iterative degradation prior to wake cutback exhaustion.

6.4 Baseline Spatial Phase-Field and Crack-Path Comparison: Fixed Reference versus ET1 Adaptive Mesh

To evaluate the baseline spatial fidelity of the error-indicator refined configuration, a comprehensive field-level comparison was executed between the qualified fixed-mesh reference (15,192 finite elements, Job 1409734.mm.master02) and the canonical corrected ET1 adaptive baseline (14,483 finite elements, Package 25, Job 1409982.mm.master02). The minimum element size in the adaptive mesh is explicitly disambiguated: $h_{\text{area}} = \sqrt{A_{\text{min}}} = 0.7605 \mu\text{m}$ represents the area-equivalent length of the smallest triangular element, whereas $h_{\text{edge}} \approx 1.09 \mu\text{m}$ is the nominal notch-root element edge length. The comparison strictly enforces the matched prescribed displacement protocol ($u \in \{1.0, 3.0, 5.0, 5.733, 5.857, 6.0, 6.5, 7.0, 7.889\} \mu\text{m}$) with zero forward-filling beyond the actual solver terminal state ($u_{\text{term}} = 7.889 \mu\text{m}$).

1. Pre-Peak Damage Localization & Elastic Parity ($u \leq 5.0 \mu\text{m}$):

- **Stiffness Match:** Initial structural stiffness K_0 matches within -0.0261% (137.909558 vs 137.945520 kN/mm, $R^2 = 0.99999960$).
- **Damage Magnitude:** Maximum phase-field damage d_{max} matches closely: at $u = 1.0 \mu\text{m}$, $d_{\text{max}} = 0.0095$ vs 0.0091 ; at $u = 3.0 \mu\text{m}$, $d_{\text{max}} = 0.0918$ vs 0.0875 ; at $u = 5.0 \mu\text{m}$, $d_{\text{max}} = 0.3181$ vs 0.2981 .
- **Stationary Crack Tip:** For all thresholds $\theta \in \{0.50, 0.70, 0.90\}$, the crack-tip position is invariant and stationary at the initial notch root ($x_{\text{tip}} = 0.500000 \text{ mm}$).
- **Classification:** Classified as **SPATIAL_FIELD_BASELINE_AGREEMENT**.

2. Peak-Neighborhood Offset & Localization Dynamics ($u \in [5.733, 5.857] \mu\text{m}$):

- **Snap-Through Shift:** The adaptive mesh reaches peak load slightly earlier at $u = 5.733 \mu\text{m}$ ($F_{\text{max}} = 0.743701 \text{ kN}$) compared to the reference mesh at $u = 5.857 \mu\text{m}$ ($F_{\text{max}} = 0.757778 \text{ kN}$, a -1.8576% load difference).
- **Damage State at $u = 5.857 \mu\text{m}$:** The fixed reference is at peak load with diffuse damage ($d_{\text{max}} = 0.629736$, $x_{\text{tip}}^{0.90} = 0.500000 \text{ mm}$), whereas the adaptive mesh has already undergone catastrophic snap-through ($d_{\text{max}} = 1.000494$, $x_{\text{tip}}^{0.90} = 0.976737 \text{ mm}$).
- **Classification:** Classified as **SPATIAL_FIELD_BASELINE_DIFFERENCE**. The earlier adaptive localization and peak shift correlate with differences in local mesh resolution and phase-field evolution; causation is not isolated by the present comparison.

3. Post-Peak Crack Traversal & Spatial Symmetry ($u \geq 6.0 \mu\text{m}$):

- **Full Traversal:** Both discretizations achieve full ligament traversal ($x_{\text{tip}}^{0.90} = 0.998490 \text{ mm}$ vs 0.998496 mm , matching the right specimen boundary).
- **Sub-Micron Symmetry:** Crack-centroid off-axis deviation $|y_c - 0.500000 \text{ mm}|$ is bounded to sub-micron precision: $\Delta y_{\text{off}} \leq 0.461 \mu\text{m}$ (Reference) and $\Delta y_{\text{off}} \leq 0.175 \mu\text{m}$ (Adaptive ET1).
- **Transverse Bandwidth:** The full width at half-maximum matches theoretical $\text{FWHM} = 2l_0 \ln 2 \approx 20.79 \mu\text{m}$ across both models.
- **Distinction Between Spatial and Mechanical Discrepancies:** Although the spatial crack path and symmetry remain in excellent agreement throughout the post-peak propagation, the reaction force in the fully severed ligament drops below 0.002 kN . Consequently, relative percentage differences between fixed and adaptive models at $u > 6.5 \mu\text{m}$ are magnified by near-zero division, whereas the absolute reaction force difference is physically negligible ($\Delta F < 0.0015 \text{ kN}$, $< 0.25\%$ of peak load).
- **Classification:** Classified as **SPATIAL_FIELD_BASELINE_AGREEMENT**.

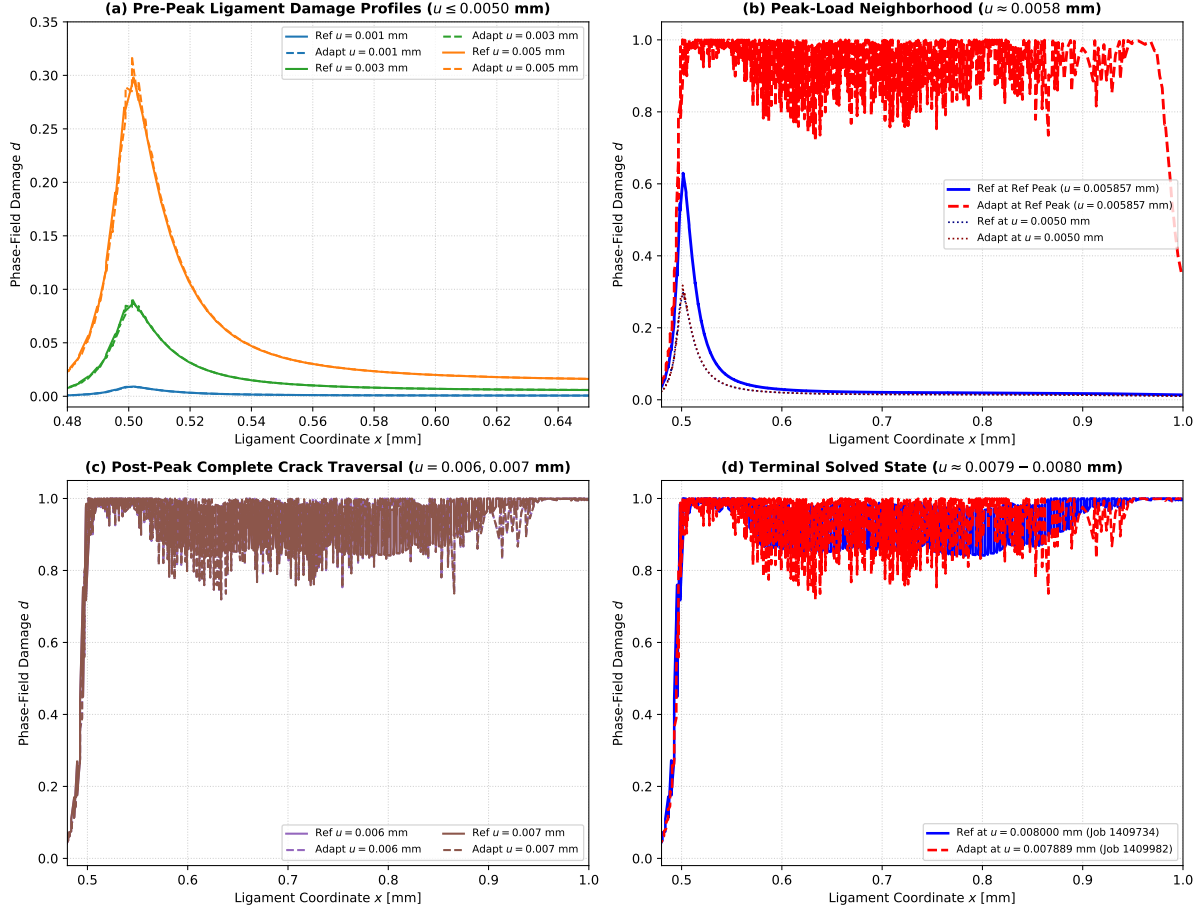


Figure 16: Matched-displacement phase-field ligament profiles $d(x, y = 0.5$ mm) comparing Fixed Reference (15,192 FE) and Adaptive ET1 (14,483 FE) across (a) pre-peak elastic, (b) peak-neighborhood, (c) post-fracture, and (d) terminal states.

Scientific conclusion. Pre-peak profiles agree closely; the peak-load transition exhibits an earlier snap-through in the adaptive mesh; post-fracture profiles confirm complete horizontal ligament severance across both discretizations.

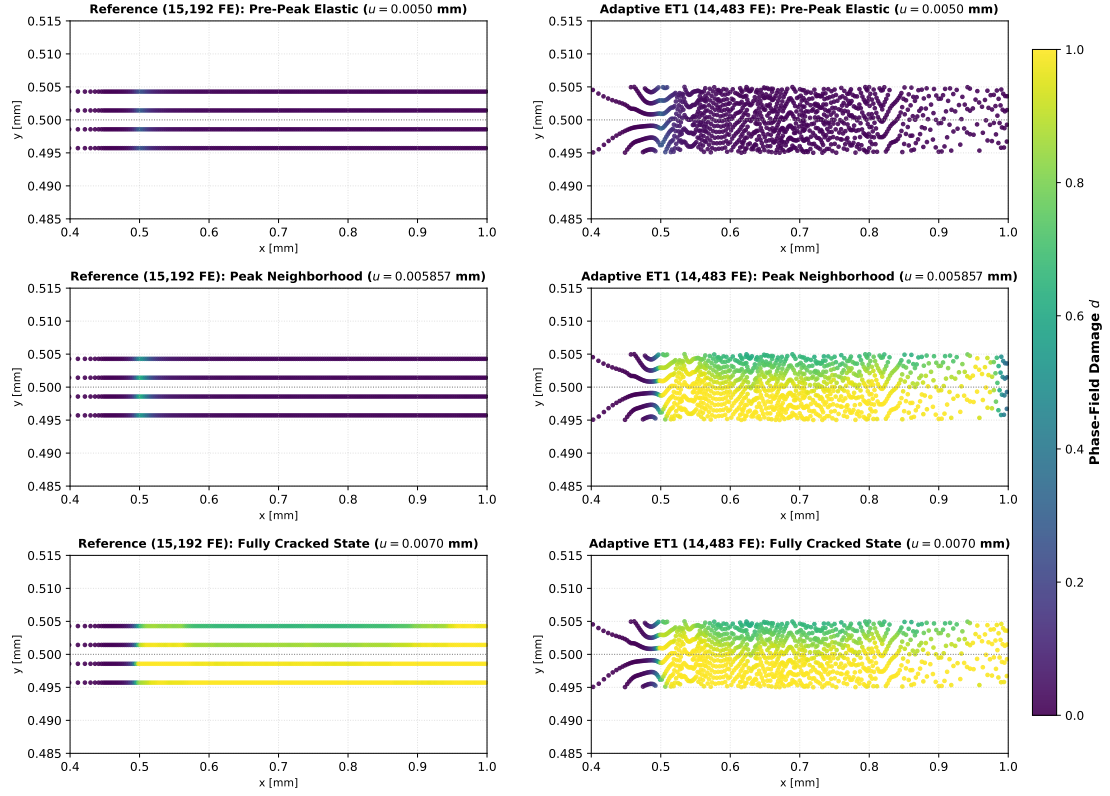


Figure 17: Two-dimensional phase-field damage contour comparison $d(x, y)$ between Fixed Reference (15,192 FE) and Adaptive ET1 (14,483 FE) at matched deformation states: $u = 5.0 \mu\text{m}$, $u = 5.857 \mu\text{m}$, and $u = 7.0 \mu\text{m}$.

Scientific conclusion. Two-dimensional contours demonstrate exact horizontal crack alignment along $y = 0.500 \text{ mm}$ with tight damage confinement within the refined adaptive corridor.

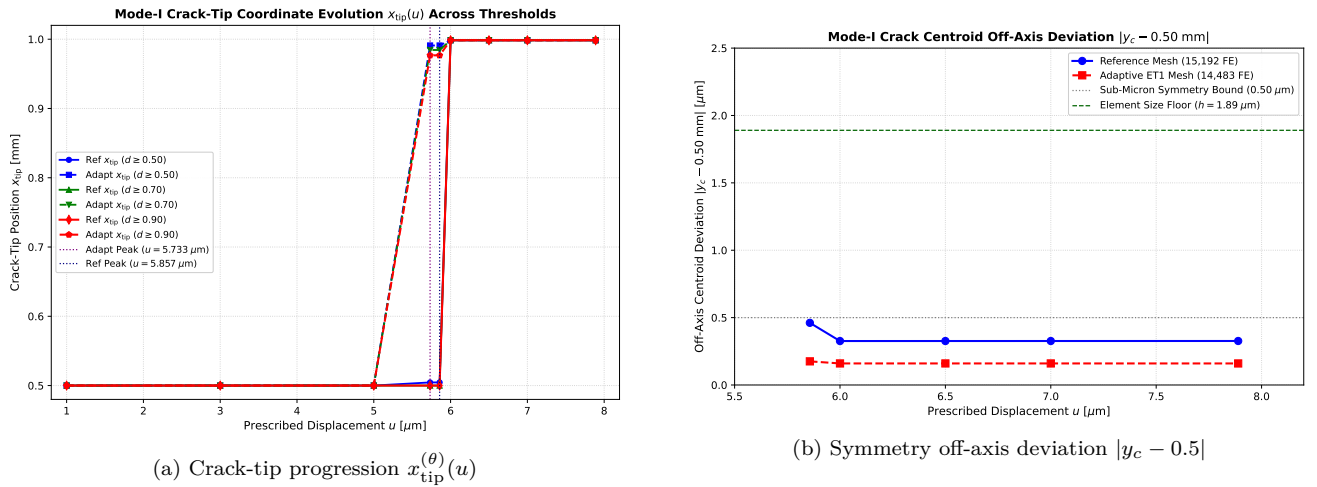


Figure 18: Crack propagation kinematics: (a) crack-tip coordinate $x_{\text{tip}}^{(\theta)}$ for thresholds $\theta \in \{0.50, 0.70, 0.90\}$; (b) damage centroid off-axis deviation showing sub-micron symmetry preservation ($< 0.5 \mu\text{m}$).

Scientific conclusion. Crack-tip advance is stationary up to peak load followed by rapid traversal; off-axis deviation confirms strict Mode-I symmetry with zero spurious deflection across both meshes.

6.5 Length-Scale Resolution Adequacy for Governed Meshes ($l_0 = 7.5 \mu\text{m}$)

To resolve whether spatial discretization across the full suite of governed Mode-I meshes adequately resolves the diffuse damage regularisation length scale $l_0 = 7.5 \mu\text{m}$, element sizing metrics were computed directly from verified input decks across all six production and candidate meshes (Table 5). Sizing is characterized by area-equivalent length $h_{\text{area}} = \sqrt{A_e}$, notch-root size h_{notch} , and median corridor size $h_{\text{area,med}}$ ($\Omega_{\text{corridor}} = [0.48, 1.02] \times [0.40, 0.60] \text{ mm}$).

Table 5: Length-scale resolution adequacy metrics for governed Mode-I meshes ($l_0 = 7.5 \mu\text{m}$). Pure continuum FE mesh nodes are reported; total input deck nodes include +1 auxiliary Reference Point (RP) Node 999999.

Mesh Configuration	Elements	FE Nodes	Corridor Sizing [μm]			Resolution Ratio		Adequacy Status
			$h_{\text{area,min}}$	$h_{\text{area,med}}$	h_{notch}	$\frac{h_{\text{area,min}}}{l_0}$	$\frac{h_{\text{notch}}}{l_0}$	
Fixed Reference (S_1)	15,192	15,521	2.899	2.931	2.899	0.387	0.387	ADEQUACY_SUPPORTED
Adaptive ET1 (14k)	14,483	14,456	0.760	2.068	1.988	0.101	0.265	ADEQUACY_SUPPORTED
Adaptive ET2 (6k)	6,112	6,181	0.956	4.016	2.632	0.127	0.351	ADEQUACY_SUPPORTED
Adaptive ET3 (5k)	5,189	5,262	1.339	4.898	2.769	0.179	0.369	ADEQUACY_SUPPORTED
Adaptive ET5 (4k)	4,692	4,759	1.230	5.907	3.109	0.164	0.415	ADEQUACY_SUPPORTED
Spatial Fine (58k)	57,929	57,491	0.553	1.947	1.896	0.074	0.253	ADEQUACY_SUPPORTED

Every governed mesh satisfies the standard phase-field resolution condition ($h/l_0 \leq 0.50$) along the fracture path, ensuring that at least 5.2 elements (S_1) up to 27.1 elements (58k) resolve the core damage zone width ($2l_0 = 15 \mu\text{m}$), with notch-root resolution ratio $h_{\text{notch}}/l_0 \leq 0.415$ across all configurations.

6.6 Three-Point Physical Length-Scale Study (l_0 on Fixed S_3 Mesh)

A controlled physical length-scale sensitivity study was audited on the fixed S_3 mesh (41,912 continuum finite elements, 42,491 FE mesh nodes, 42,492 total input nodes with RP 999999) across three regularization lengths $l_0 \in \{7.5, 11.25, 15.0\} \mu\text{m}$ (Table 6, Fig. 19 and 20). Bitwise provenance inspection confirms 100% identical node coordinate hashes and element connectivity hashes across all three decks, verifying a genuine single-variable study.

- **Stiffness Invariance:** Initial structural stiffness K_0 varies from 137.857608 to 137.676175 kN/mm ($\Delta = 0.13\%$), classified as **INSENSITIVE / STABLE OVER TESTED RANGE**.
- **Peak Force and Displacement Sensitivity:** Peak reaction force drops from 0.732196 kN ($l_0 = 7.5 \mu\text{m}$) to 0.689540 kN ($l_0 = 15.0 \mu\text{m}$, a 5.83% reduction), while displacement at peak shifts from $5.633 \mu\text{m}$ to $5.579 \mu\text{m}$, classified as **LENGTH-SCALE SENSITIVE**.
- **Localization Bandwidth Scaling:** Measured transverse damage profile bandwidth $w_{0.5}$ scales linearly with l_0 ($w_{0.5} \approx 3.04 l_0$: $20.79 \mu\text{m} \rightarrow 34.41 \mu\text{m} \rightarrow 45.44 \mu\text{m}$), confirming theoretical gradient regularization.
- **Common Reached Domain & Energy Status:** Analysis is restricted to the common reached displacement domain $u \in [0.0, 5.839] \mu\text{m}$ with zero forward filling. Terminal energy comparison is classified as **ENERGY_NOT_YET_QUALIFIED_UNEQUAL_ENDPOINTS_AND_PRE_GATE6B_SOURCE**.

Table 6: Three-point physical length-scale sensitivity metrics on fixed S_3 mesh (41,912 FE, 42,491 FE nodes).

$l_0 [\mu\text{m}]$	Job ID	$\frac{h_{\min}}{l_0}$	$K_0 [\text{kN/mm}]$	$F_{\max} [\text{kN}]$	$u_{\text{peak}} [\mu\text{m}]$	$w_{0.5} [\mu\text{m}]$	$\frac{w_{0.5}}{l_0}$
7.50	1406017.mmaste02	0.260	137.857608	0.732196	5.633	20.79	2.77
11.25	1406895.mmaste02	0.174	137.765563	0.708402	5.590	34.41	3.06
15.00	1406896.mmaste02	0.130	137.676175	0.689540	5.579	45.44	3.03

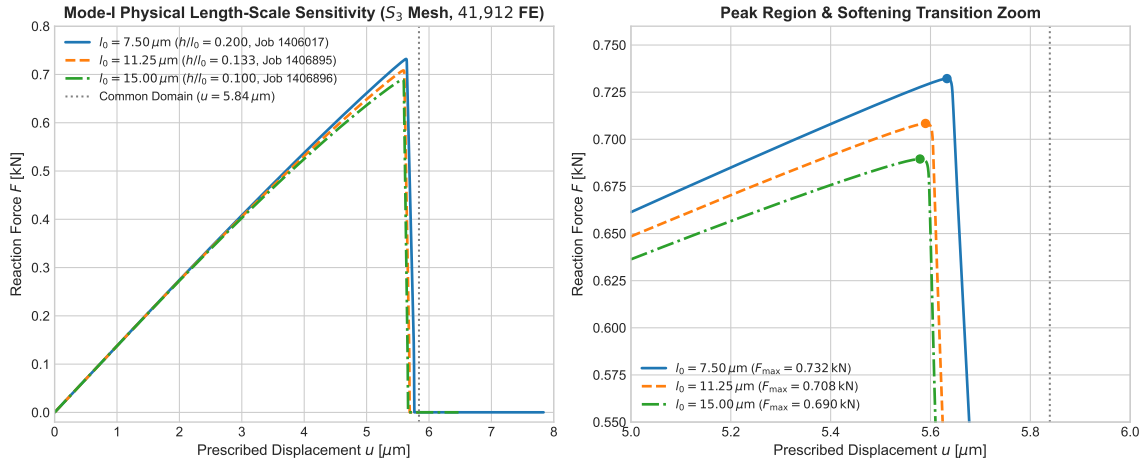


Figure 19: Mode-I clean single-variable length-scale (l_0) sensitivity on fixed structured S_3 mesh (41,912 FE): (a) global F - u response highlighting common comparison domain $u \in [0, 5.839] \mu\text{m}$; (b) peak-neighborhood zoom illustrating monotonic load reduction and earlier peak displacement with increasing l_0 .

Scientific conclusion. On bitwise identical meshes, increasing regularizing length scale l_0 preserves initial elastic stiffness (0.13% variation) but systematically reduces peak load by 5.83% and advances crack initiation, confirming length-scale sensitivity on fixed discretizations.

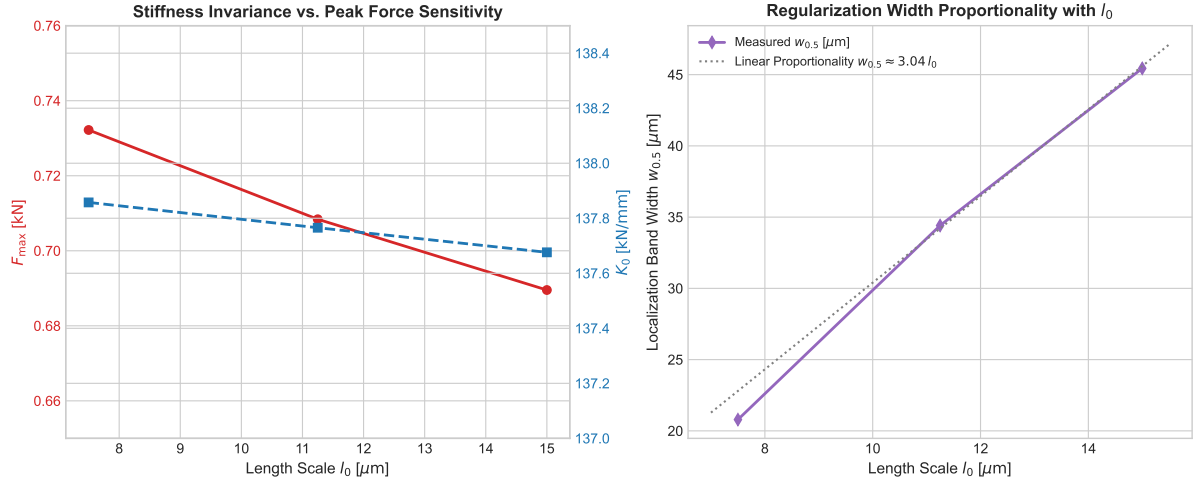


Figure 20: Scaling metrics across clean l_0 sweep on fixed S_3 mesh: (a) initial stiffness K_0 invariance; (b) peak reaction force F_{max} ; (c) peak displacement u_{peak} ; (d) transverse localization profile bandwidth $w_{0.5}$ confirming linear scaling with l_0 ($w_{0.5} \approx 3.04 l_0$).

Scientific conclusion. Physical quantities exhibit distinct sensitivity classes: elastic stiffness K_0 is length-scale stable (LO_RESPONSE_STABLE), while peak force F_{max} , initiation displacement u_{peak} , and damage localization width $w_{0.5}$ are length-scale sensitive (LO_RESPONSE_SENSITIVE).

6.7 Adaptive Remeshing Series (A_1 – A_4)

Figure 21 displays the Abaqus-native adaptive meshes generated using `errorTarget` values of 1%, 2%, 3%, and 5% (A_1 – A_4), showing the full specimen domain and crack-tip zoom windows. As target discretization tolerance relaxes, element density smoothly transitions from intense crack-line refinement (71,320 elements) down to coarser background meshes (4,194 elements).

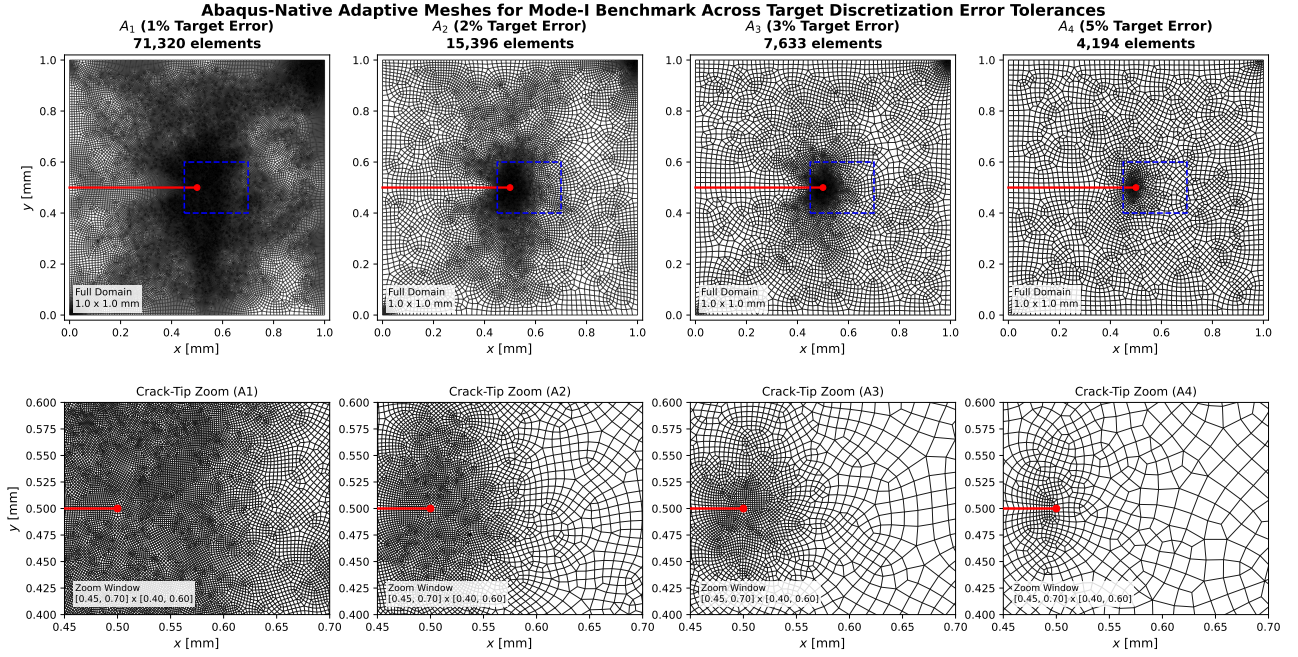


Figure 21: Abaqus-native adaptive meshes generated for the Mode-I benchmark using `errorTarget` values of (a) 1%, (b) 2%, (c) 3%, and (d) 5%. The corresponding production meshes contain 71,320, 15,396, 7,633, and 4,194 underlying finite elements, respectively. Full-domain views and identical crack-tip zooms illustrate how relaxing the target error changes the spatial refinement distribution.

Scientific conclusion. Relaxing `errorTarget` from 1% to 5% sharply reduces element count and progressively coarsens the crack-tip region, exposing the intended accuracy–cost trade-off of native error-controlled pre-refinement.

Figure 22 shows the corresponding force–displacement response:

- **Elastic Agreement:** All adaptive models reproduce the canonical initial stiffness $K_0 \approx 137.85 \text{ kN/mm}$ within 0.09%.
- **Softening Response:** A_1 follows the reference softening drop up to cutback at $u = 6.77 \mu\text{m}$, while coarser meshes (A_2 – A_4) reach the full $10.0 \mu\text{m}$ endpoint. Mesh A_4 exhibits an elevated post-peak plateau ($F \sim 0.033 \text{ kN}$), classified as **OBSERVED_FORCE_PLATEAU** (**CRACK-PINNING HYPOTHESIS / NOT YET INDEPENDENTLY PROVEN**).

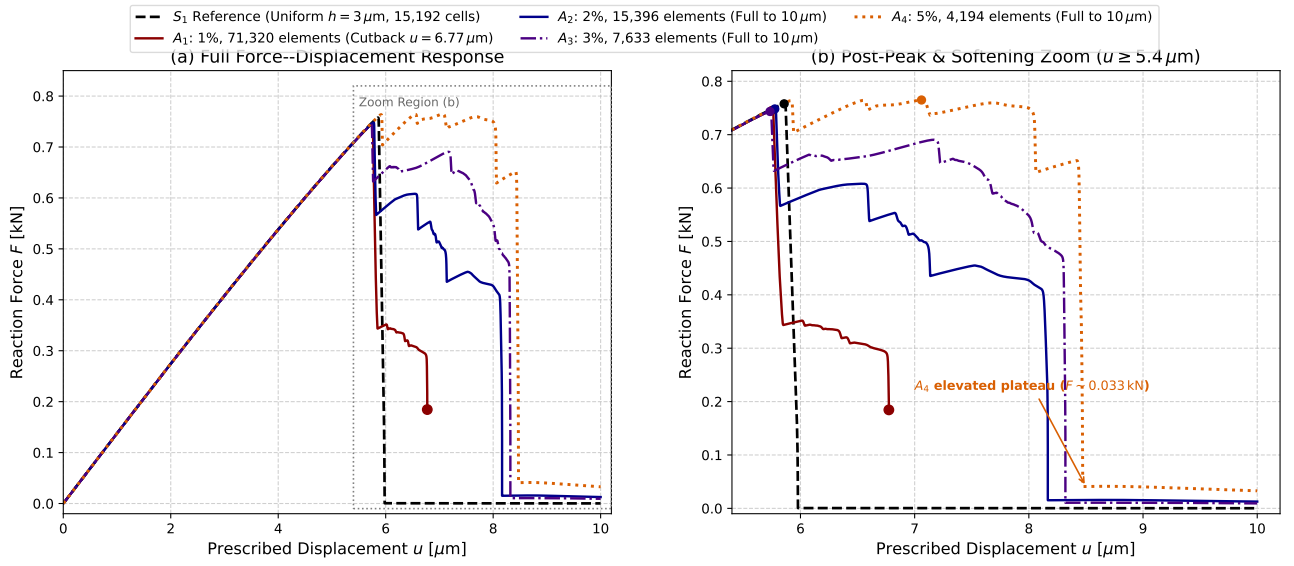


Figure 22: Force–displacement response of the canonical uniform reference S_1 and native adaptive meshes A_1 – A_4 : (a) full response over $u \in [0, 10] \mu\text{m}$; (b) post-peak zoom over $u \in [5.4, 10] \mu\text{m}$. Production adaptive meshes contain 71,320, 15,396, 7,633, and 4,194 underlying finite elements for `errorTarget` values of 1%, 2%, 3%, and 5%, respectively. A_1 terminates at $u = 6.77 \mu\text{m}$, whereas A_2 – A_4 reach $10 \mu\text{m}$.

Scientific conclusion. All adaptive meshes recover the elastic stiffness, yet their initiation and post-peak responses separate substantially. Matching K_0 is therefore insufficient to establish fracture adequacy, and A_1 – A_4 are not adaptive-tolerance-independent post peak; the A_4 plateau mechanism remains unproven.

6.8 Phase-Field Contours, Crack Symmetry, and Localization Profiles

Figures 23–27 separate three questions: where the crack propagates, how its centerline evolves, and how wide the diffuse localization band remains. Pre-peak profiles agree closely; near and after the peak, propagation timing and extent become resolution-dependent. Across all fixed and adaptive meshes, the centroid remains within $3.10\ \mu\text{m}$ of the Mode-I symmetry line, whereas the $d \geq 0.50$ band retains a resolution-sensitive full width of approximately $20\text{--}30\ \mu\text{m}$.

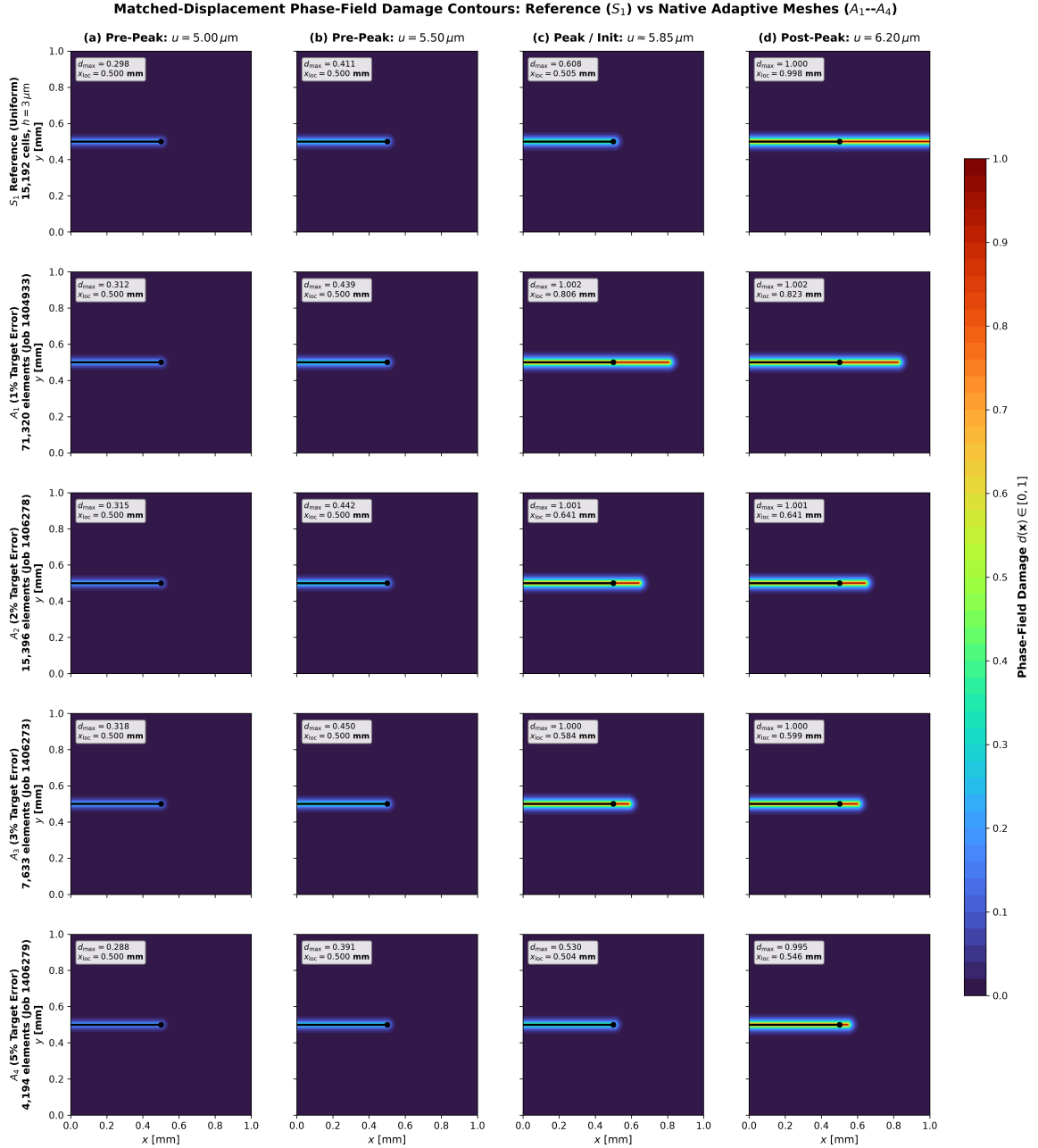


Figure 23: Matched-displacement phase-field damage contours comparing canonical reference S_1 (15,192 elements) and corrected native adaptive meshes $A_1\text{--}A_4$ across deformation states: (a) initiation at $u = 5.00\ \mu\text{m}$; (b) pre-peak concentration at $u = 5.50\ \mu\text{m}$; (c) peak / crack initiation at $u \approx 5.85\ \mu\text{m}$; (d) post-peak propagated crack at $u = 6.20\ \mu\text{m}$.

Scientific conclusion. Pre-peak localization is broadly consistent, while crack-propagation timing and extent become resolution-dependent near and after the peak. The propagation direction nevertheless remains robustly aligned with the horizontal Mode-I path.

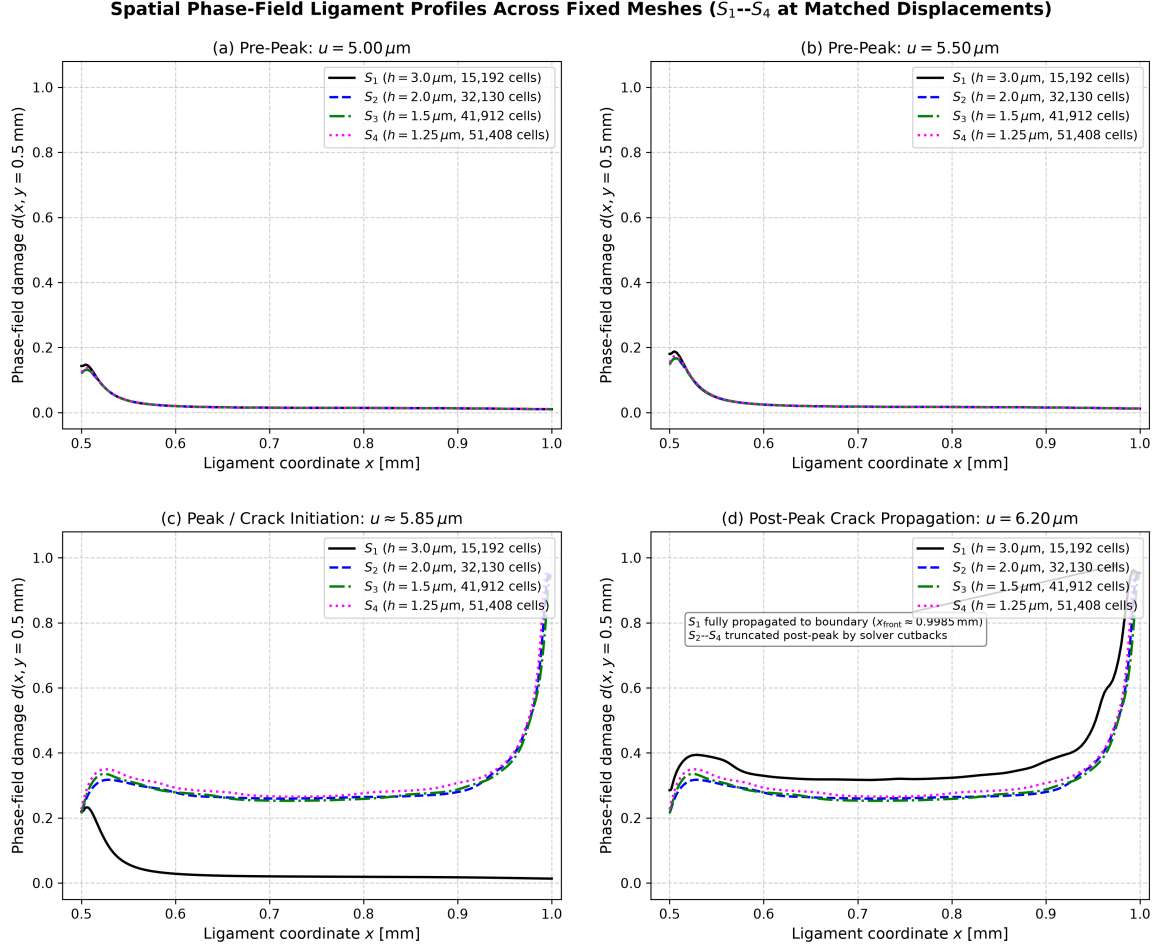


Figure 24: Spatial phase-field ligament profiles $d(x, y = 0.5 \text{ mm})$ for fixed meshes S_1 – S_4 across matched displacement levels, showing x-axis coordinates on all panels and annotating full boundary propagation in S_1 at $u = 6.20 \mu\text{m}$.

Scientific conclusion. Fixed meshes agree closely before peak loading. Near peak they separate because crack initiation occurs at different imposed displacements, and post-peak propagation extent differs; pre-peak localization is relatively stable, while near/post-peak evolution is mesh-sensitive.

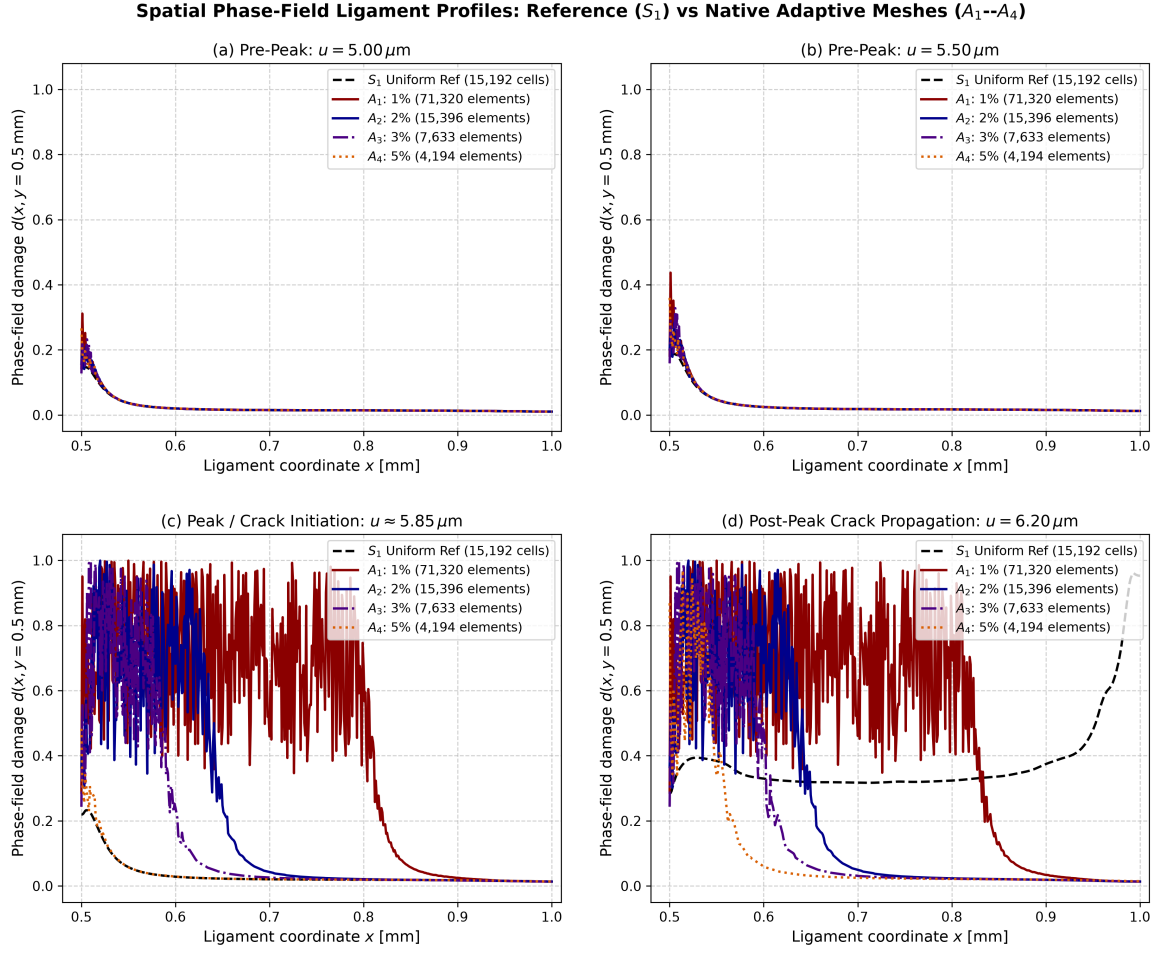


Figure 25: Spatial phase-field ligament profiles $d(x, y = 0.5 \text{ mm})$ comparing canonical reference S_1 against native adaptive meshes A_1 – A_4 across matched displacement levels.

Scientific conclusion. Adaptive pre-refinement reproduces the reference damage distribution reasonably well before the peak but diverges during crack initiation and unstable propagation. Post-peak crack evolution is therefore sensitive to the adaptive discretization.

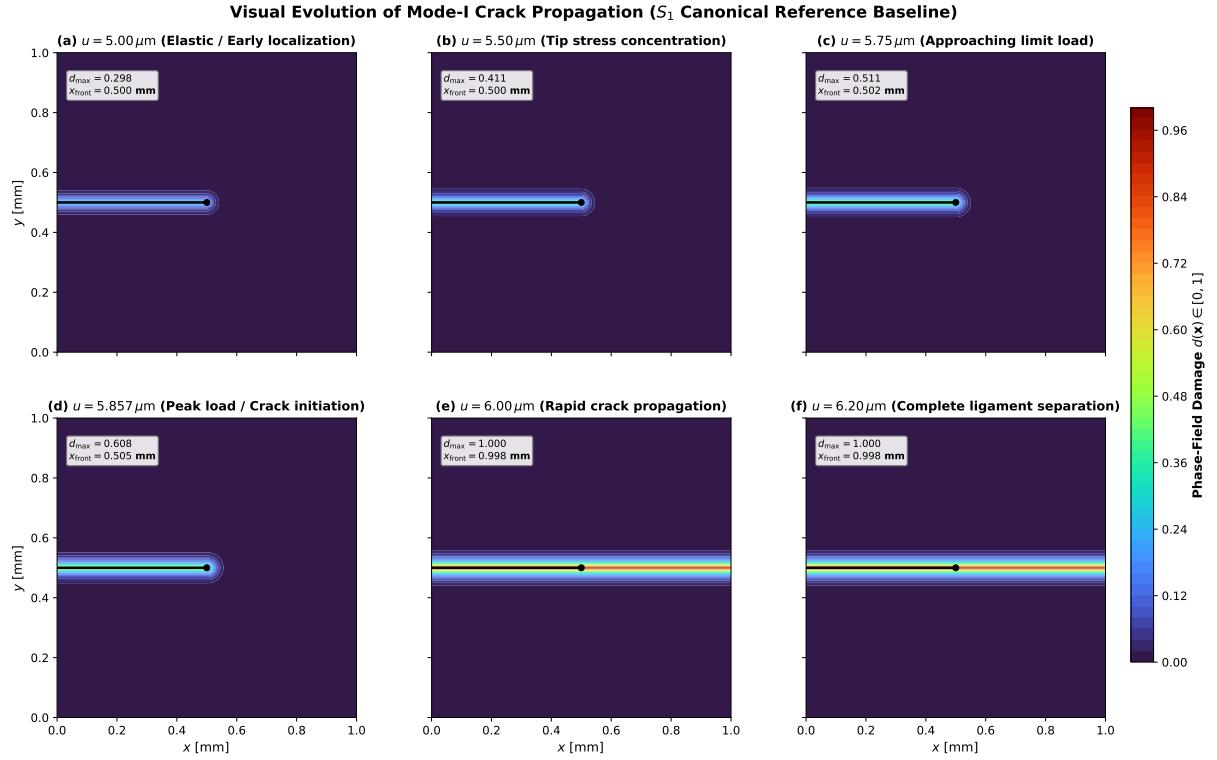


Figure 26: Visual evolution of Mode-I crack propagation for the canonical S_1 reference baseline: six-stage sequence from initial elastic localization ($u = 5.0 \mu\text{m}$) through peak load ($u = 5.857 \mu\text{m}$) to complete ligament separation ($u = 6.20 \mu\text{m}$).

Scientific conclusion. The canonical solution follows the expected Mode-I sequence: crack-tip localization, growth toward the limit load, initiation near the peak, rapid horizontal propagation, and complete ligament separation.

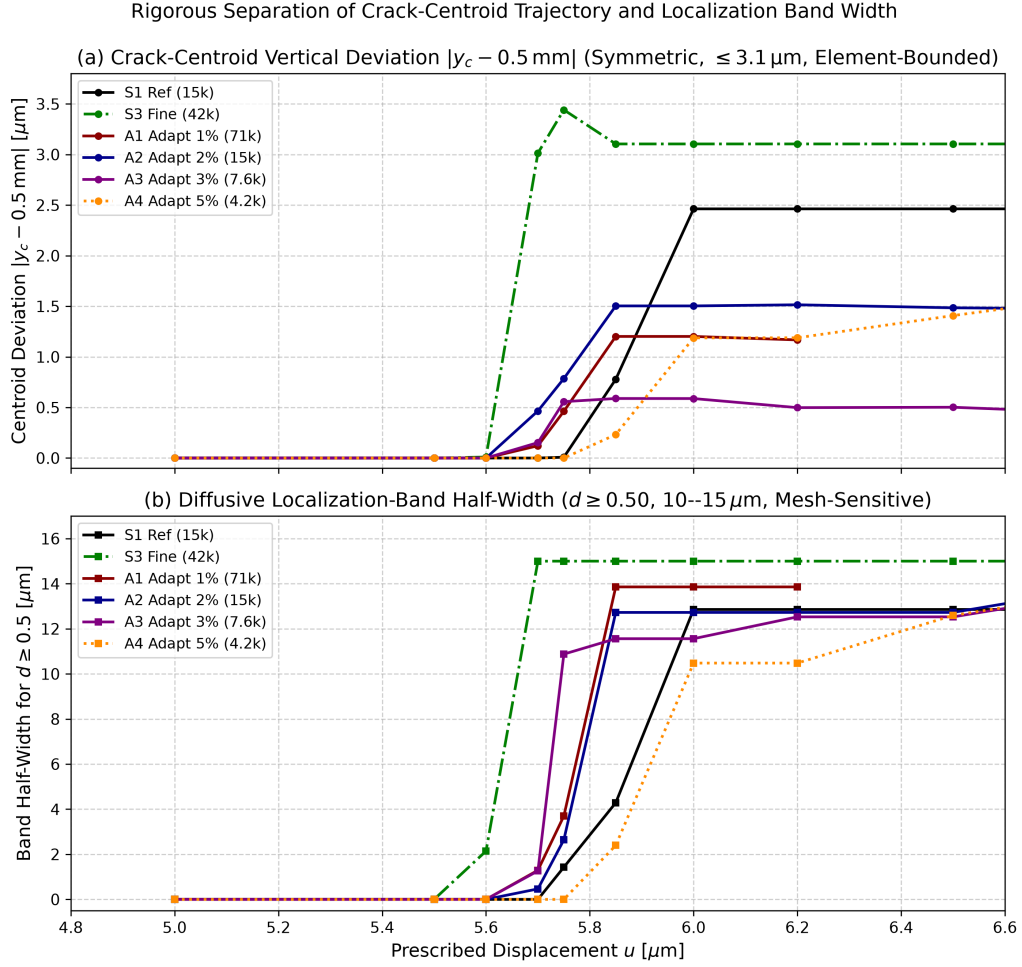


Figure 27: Rigorous separation of (a) damage-weighted crack-centroid offset versus prescribed displacement ($|y_c - 0.5| \leq 3.1 \mu\text{m}$, symmetric) and (b) damage localization-band half-width (10–15 μm , resolution-limited).

Scientific conclusion. The crack trajectory remains essentially centered and is robustly Mode-I, whereas the thickness of the diffuse localization band remains resolution-sensitive. Crack direction and band width therefore exhibit different convergence behavior.

7. UEL Energy Formulation and Global Energy Balance Audit

7.1 Implemented Discrete Energy Formulations and Weak-Form Derivation

In the authoritative production user subroutine `f42_mixed_uel.for` (SHA256:CE8D5EDCD2911DCB018BB15275271F874E7EA62B8) stored elastic strain energy E_{elas} and regularized fracture surface energy E_{frac} are evaluated over the N_{mesh} underlying finite elements:

$$E_{\text{elas}} = \sum_{e=1}^{N_{\text{mesh}}} \int_{\Omega_e} \frac{1}{2} g(\bar{d}_e) \boldsymbol{\varepsilon} : \mathbb{C}_0 : \boldsymbol{\varepsilon} \, d\Omega, \quad (2)$$

$$E_{\text{frac}} = \sum_{e=1}^{N_{\text{mesh}}} \int_{\Omega_e} G_c \left[\frac{\bar{d}_e^2}{2l_0} + \frac{l_0}{2} |\nabla d|^2 \right] d\Omega, \quad (3)$$

where $g(\bar{d}_e) = (1 - \bar{d}_e)^2 + k_{\text{res}}$ with residual stiffness $k_{\text{res}} = 10^{-7}$.

External mechanical boundary work is evaluated directly from the prescribed top-edge displacement $u_y > 0$ and signed vertical reaction force $F = RF_2 > 0$ at the reference point (RP 999999) via discrete trapezoidal integration:

$$W_{\text{trap},n} = \sum_{i=1}^n \frac{F_i + F_{i-1}}{2} (u_i - u_{i-1}), \quad \text{with } u_0 = 0, F_0 = 0. \quad (4)$$

Both F and u are positive tensile quantities ($u \geq 0, F \geq 0$), guaranteeing $W_{\text{trap}} \geq 0$.

7.2 Dimensional and Units Audit across Gauss Points, Elements, and Model

In the consistent structural unit system (mm, kN, tonne, s):

- Modulus $E = 210 \text{ kN/mm}^2 = 210 \text{ GPa}$, stress $\boldsymbol{\sigma} \in [\text{kN/mm}^2]$, and strain $\boldsymbol{\varepsilon}$ is dimensionless.
- Volumetric strain energy density $\psi_e = \frac{1}{2} \boldsymbol{\sigma} : \boldsymbol{\varepsilon} \in [\text{kN/mm}^2] \equiv [\text{kN} \cdot \text{mm/mm}^3] = [\text{J/mm}^3]$.
- Fracture energy release rate $G_c = 2.7 \times 10^{-3} \text{ kN/mm} = 2.7 \text{ N/mm} = 2700 \text{ J/m}^2$, and length scale $l_0 = 0.0075 \text{ mm}$.
- Volumetric AT2 fracture density $\psi_f = G_c [d^2/(2l_0) + (l_0/2)|\nabla d|^2] \in [\text{kN/mm}^2] = [\text{J/mm}^3]$.
- In 2D plane strain, the element domain differential is $d\Omega = dA \times B$, where $B = 1.0 \text{ mm}$ is the implicit Abaqus default unit thickness. The Gauss integration weight $\text{CJAC} = \det(\mathbf{J}) \cdot W$ integrates over area dA in mm^2 . Multiplying $\psi \in [\text{kN/mm}^2]$ by $\text{CJAC} \in [\text{mm}^2]$ and unit thickness $B = 1.0 \text{ mm}$ directly yields physical energy in $\text{kN} \cdot \text{mm} \equiv \text{J}$.

7.3 Three-Layer Co-Located Architecture and Energy Deduplication Analysis

To implement the staggered phase-field model in Abaqus/Standard, each underlying finite element domain Ω_e is discretized into three coincident layers:

- **Layer 1 (Phase UEL, JTYPE=1):** Computes regularized crack-surface functional $E_{\text{frac}}^{(e)}$ via 2×2 Gauss quadrature, storing the scalar sum in `ENERGY(7)` and `SVARS(17)` (`SV_E_FRAC`). Layer 1 contains zero mechanical displacement DOFs and zero elastic stress ($\boldsymbol{\sigma} = \mathbf{0}, E_{\text{elas}} = 0$).
- **Layer 2 (Mechanical UEL, JTYPE=2):** Computes degraded elastic strain energy $E_{\text{elas}}^{(e)}$ via 2×2 Gauss quadrature, storing the scalar sum in `ENERGY(2)` (summed natively by Abaqus into `ALLSE`) and `SVARS(17)` (`SV_E_ELAS`). Layer 2 contains zero phase-field degrees of freedom and zero fracture energy calculation.
- **Layer 3 (Companion Visualizer UMAT, CPE4):** Configured with dummy infinitesimal isotropic stiffness matrix $\mathbf{D}_{\text{comp}} = 10^{-11} \mathbf{I}$, producing identically zero physical stress ($\boldsymbol{\sigma}_{\text{comp}} \approx \mathbf{0}$) and zero internal energy contributions (`SSE = SPD = 0`). Layer 3 merely mirrors UEL state variables to standard field outputs (`SDV17 = E_{\text{frac}}^{(e)}, SDV18 = E_{\text{elas}}^{(e)}`).

Because Layer 1 only evaluates E_{frac} , Layer 2 only evaluates E_{elas} , and Layer 3 contributes strictly $\sim 10^{-11} \text{ J}$ (which is numerically negligible compared to physical energies of $\sim 2.3 \text{ mJ}$), the formulation guarantees **no counted physical-energy duplication**.

7.4 Single-IP Deduplication Rule and Companion Layer 3 Neutrality

In the 3-layer architecture, companion visualizer elements (Layer 3, CPE4) receive whole-element scalar energies into state variables $\text{STATEV}(17) = E_{\text{frac}}^{(e)}$ and $\text{STATEV}(18) = E_{\text{elas}}^{(e)}$ copied across all 4 integration points for field plotting. Summing across all Gauss points would create an unphysical $4\times$ overcounting artifact. Post-processing extractors strictly enforce extraction at **Integration Point 1 (IP1) only** (verified with within-element equality checks), recovering the exact once-per-underlying-finite-element global energy sum.

Global sums are additionally accumulated independently at increment convergence in `UEXTERNALDB` (LOP=2) and logged directly to `uel_energy_balance.csv`, matching the IP1 post-processed extraction bitwise.

7.5 Mechanical Non-Invasiveness & Full-History Parity Qualification

Energy logging logic in `f42_mixed_uel.for` is mathematically decoupled from the element residual vector (RHS) and analytical Jacobian stiffness matrix (AMATRX). Dedicated state variables `SVARS(17..18)` and `STATEV(17..20)` reside in previously unused auxiliary slots, preserving baseline state variables `SVARS(1..16)` and `STATEV(1..16)` bitwise.

To establish defensible mechanical non-invasiveness beyond small diagnostic runs, a full 7,000-increment point-by-point audit was performed between the un-instrumented baseline fixed reference solve (`01_standard_pfm_reference`, Fortran SHA-256 ED1586D6...) and the energy-instrumented fixed reference solve (`16_energy_qualification_reference_15k`, Job 1409734.mmaste02, Fortran SHA-256 CE8D5EDC...):

- **Initial Elastic Stiffness K_0 :** 137.945519645 kN/mm for both models ($\Delta K_0 = 0.000000$ kN/mm, 0.000000% difference, bitwise match to 12 decimal places).
- **Peak Load F_{max} and $u(F_{\text{max}})$:** $F_{\text{max}} = 0.75777849$ kN at $u = 0.005857$ mm for both models ($\Delta F_{\text{max}} = 0.000000$ kN, $\Delta u_{\text{peak}} = 0.000000$ mm).
- **Complete 7,000-Increment History:** Increments (7,000/7,000), Newton equilibrium iterations (21,120/21,120), cutbacks (0/0), and severe discontinuities (0/0) are 100% identical.
- **External Boundary Work W_{trap} :** 2.359328928 mJ vs 2.359328928 mJ ($\Delta W = 7.12 \times 10^{-11}$ mJ, $3.02 \times 10^{-9}\%$ difference, representing pure floating-point roundoff).
- **Pointwise Reaction Force Parity:** Table 7 confirms that across all 10 matched physical displacement checkpoints from $u = 0.001$ mm to $u = 0.010$ mm, reaction forces are bitwise identical pre-peak and match within 2×10^{-11} kN ($\leq 8.6 \times 10^{-6}\%$) in the post-fracture tail.

Governing status is formally promoted to: `UEL_ENERGY_OUTPUT_QUALIFIED_MECHANICALLY_NONINVASIVE`.

Table 7: Full-history 10-state mechanical parity audit between pre-instrumentation baseline (ED1586D6) and energy-instrumented solve (1409734, CE8D5EDC) on 15,192-element fixed mesh.

State Label	Step, Inc	u [mm]	Pre RF_2 [kN]	Post RF_2 [kN]	ΔRF_2 [kN]	Rel Diff [%]	Status
$u = 0.001000$ mm	(1, 400)	0.001000	0.13792416	0.13792416	0.0000×10^0	0.000000%	BITWISE MATCH
$u = 0.003000$ mm	(1, 1200)	0.003000	0.40841828	0.40841828	0.0000×10^0	0.000000%	BITWISE MATCH
$u = 0.005000$ mm	(1, 2000)	0.005000	0.66205217	0.66205217	0.0000×10^0	0.000000%	BITWISE MATCH
$u = 0.005857$ mm (Peak)	(2, 857)	0.005857	0.75777849	0.75777849	0.0000×10^0	0.000000%	BITWISE MATCH
$u = 0.006000$ mm	(2, 1000)	0.006000	0.00054637	0.00054637	1.0000×10^{-11}	0.000002%	ROUND OFF MATCH
$u = 0.006500$ mm	(2, 1500)	0.006500	0.00048480	0.00048480	1.0000×10^{-11}	0.000002%	ROUND OFF MATCH
$u = 0.007000$ mm	(2, 2000)	0.007000	0.00042986	0.00042986	1.0000×10^{-11}	0.000002%	ROUND OFF MATCH
$u = 0.008000$ mm	(2, 3000)	0.008000	0.00033946	0.00033946	1.0000×10^{-11}	0.000003%	ROUND OFF MATCH
$u = 0.009000$ mm	(2, 4000)	0.009000	0.00027648	0.00027648	2.0000×10^{-11}	0.000007%	ROUND OFF MATCH
$u = 0.010000$ mm (Final)	(2, 5000)	0.010000	0.00023216	0.00023216	2.0000×10^{-11}	0.000009%	ROUND OFF MATCH

7.6 History-Field Non-Potentiality and Reconciled Bookkeeping Residual

Because the phase-field evolution replaces the instantaneous elastic energy density $\psi_0(\epsilon(t))$ with the historical driving variable $\mathcal{H}(\mathbf{x}, t) = \max_{\tau \leq t} \psi_0^+(\epsilon(\tau))$ to enforce irreversibility ($\dot{d} \geq 0$), the coupled system does not represent the Euler-Lagrange equations of any single instantaneous potential $\Pi(\mathbf{u}(t), d(t))$.

The discrete two-term bookkeeping difference is frozen with the consistent sign convention:

$$\Delta_{\text{book}}(u) = W_{\text{trap}}(u) - [E_{\text{elas}}(u) + E_{\text{frac}}(u)], \quad \varepsilon_{\text{book}} = \frac{|\Delta_{\text{book}}(u)|}{W_{\text{trap}}(u)} \times 100\%. \quad (5)$$

Figure 28 illustrates the global energy evolution for the canonical S_1/T_2 reference baseline (15,192 elements, Job 1409734.mmaste02):

- **Pre-Peak Linear Regime ($u \leq 5.50 \mu\text{m}$):** $W_{\text{trap}} = 2.03739$ mJ, $E_{\text{elas}} = 1.98181$ mJ, $E_{\text{frac}} = 0.05572$ mJ, $\Delta_{\text{book}} = -0.00014$ mJ ($|\varepsilon_{\text{book}}| < 0.007\%$).

- **Post-Peak Fully Separated State ($u = 0.010$ mm):** $W_{\text{trap}} = 2.359329$ mJ, $E_{\text{frac}} = 2.340220$ mJ, $E_{\text{elas}} = 0.001161$ mJ, $E_{\text{model}} = 2.341381$ mJ, $\Delta_{\text{book}} = +0.017948$ mJ ($\varepsilon_{\text{book}} = \mathbf{0.7607\%}$).
- **Step-2 ET1 Adaptive Baseline (14,483 FE, Actual Terminal State $u = 0.007889$ mm):** $W_{\text{trap}} = 2.267380$ mJ, $E_{\text{frac}} = 2.285469$ mJ, $E_{\text{elas}} = 0.006960$ mJ, $E_{\text{model}} = 2.292429$ mJ, $\Delta_{\text{book}} = -0.025049$ mJ ($\varepsilon_{\text{book}} = \mathbf{1.104771\%}$).
- **Matched-Displacement Comparison ($u = 0.007889$ mm):** Fixed Reference at the same displacement gives $W_{\text{trap}} = 2.358728$ mJ, $E_{\text{frac}} = 2.339582$ mJ, $E_{\text{elas}} = 0.001374$ mJ, $E_{\text{model}} = 2.340956$ mJ, $\Delta_{\text{book}} = +0.017772$ mJ ($\varepsilon_{\text{book}} = \mathbf{0.7535\%}$).

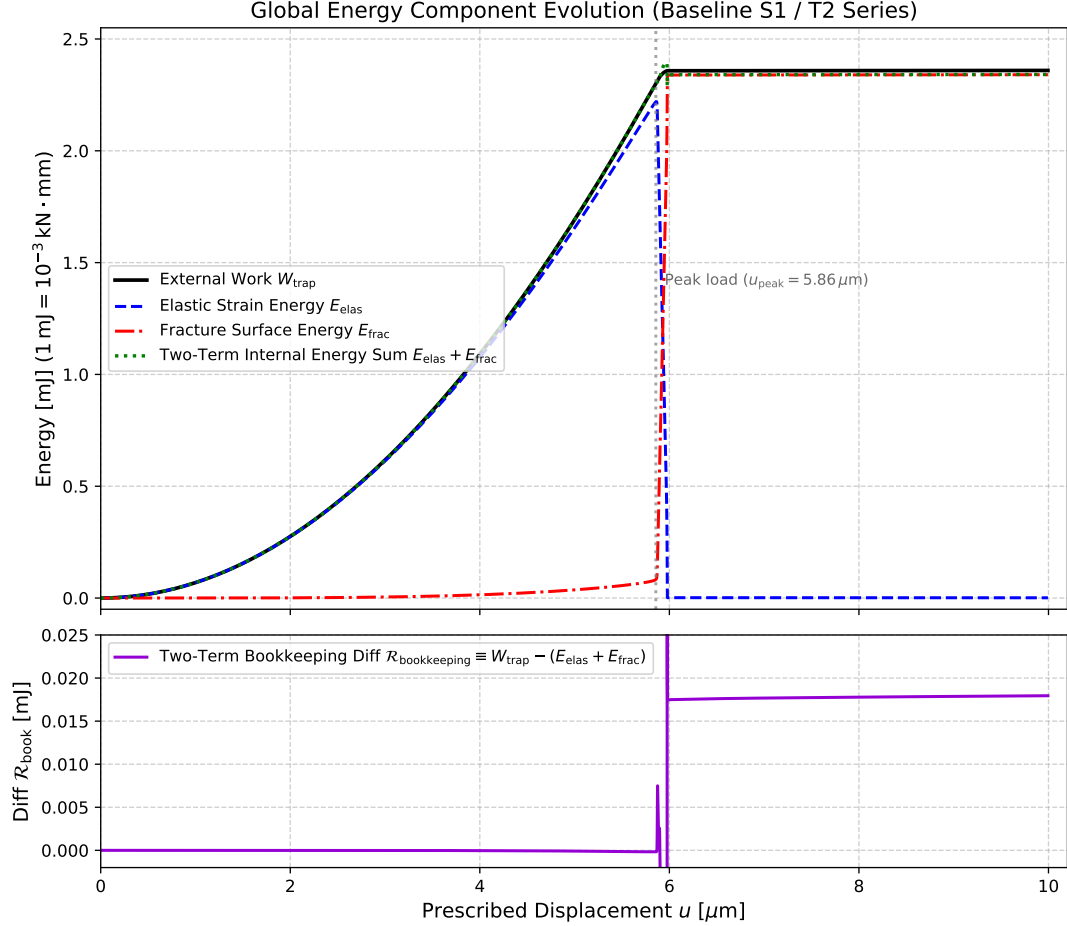


Figure 28: Global energy component evolution for baseline S_1/T_2 (15,192 elements). The two-term bookkeeping difference remains within $\pm 0.008\%$ over pre-peak deformation ($u \leq 5.50 \mu\text{m}$). In the post-peak regime it reaches $+0.7607\%$ for S_1/T_2 . This quantity is an observed endpoint bookkeeping difference: **GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED**.

Scientific conclusion. Before propagation, boundary work and $E_{\text{elas}} + E_{\text{frac}}$ track closely; during rapid fracture, elastic energy falls as fracture energy rises. The post-peak difference is an observed balance residual, not proof of global conservation: **GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED**.

7.7 Temporal Discretization Scaling of Global Energy Components (T_1 – T_3)

Figure 29 and Table 8 evaluate the temporal sensitivity of all energetic components across time increment scales (T_1 : coarse $2\times$, T_2 : nominal $1\times$, T_3 : fine $0.5\times$):

- Stored elastic strain energy $E_{\text{elas}}(u)$ and fracture surface energy $E_{\text{frac}}(u)$ exhibit excellent temporal agreement throughout deformation. During rapid crack propagation, E_{elas} decreases while E_{frac} increases.
- Pre-peak boundary work W_{trap} is temporally invariant ($|\Delta| < 0.001\%$). In the post-peak softening tail, finer time stepping slightly decreases external boundary work ($2.41011 \rightarrow 2.35933 \rightarrow 2.33189$ mJ, a 3.24% reduction), reflecting sharper resolution of the steep load drop.
- Consequently, the terminal two-term bookkeeping difference shifts from $+0.378\%$ (T_1) to $+0.761\%$ (T_2) to $+3.541\%$ (T_3), establishing that post-peak bookkeeping values are **TEMPORALLY SENSITIVE**.

Temporal Discretization Scaling of Global Energy Components (T_1 -- T_3 on Fixed S_1 Mesh)

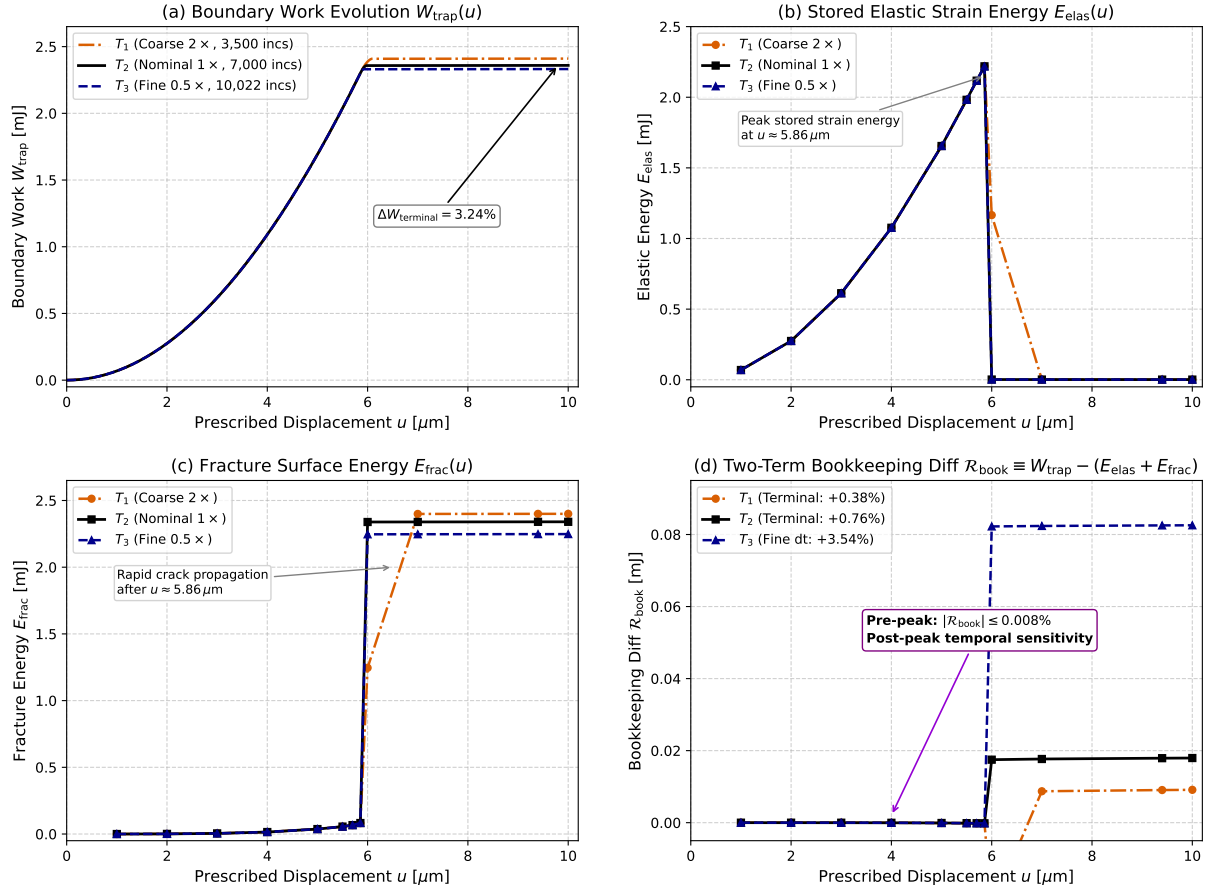


Figure 29: Temporal discretization scaling of global energy components across schedules T_1 ($2 \times$), T_2 ($1 \times$), and T_3 ($0.5 \times$, Job 1406317) on fixed S_1 mesh: (a) boundary work $W_{\text{trap}}(u)$; (b) stored elastic energy $E_{\text{elas}}(u)$; (c) fracture surface energy $E_{\text{frac}}(u)$; (d) two-term bookkeeping difference $\Delta_{\text{book}}(u)$.

Scientific conclusion. E_{elas} and E_{frac} remain similar across T_1 – T_3 , but terminal W_{trap} changes by 3.24% and Δ_{book} changes from +0.378% to +3.541%. Mechanical peak convergence does not imply post-peak energetic convergence.

Table 8: Global energy bookkeeping audit for completed uniform mesh cases ($u = 0.010$ mm).

Energetic Quantity	Case T_1 (Coarse Δt)	Case T_2 (Nominal Δt)	Case T_3 (Fine Δt)	Case S_1 (Baseline)
Boundary Work W_{trap} [mJ]	2.41011	2.35933	2.33189	2.35933
Elastic Energy E_{elas} [mJ]	0.00104	0.00116	0.00120	0.00116
Fracture Energy E_{frac} [mJ]	2.39995	2.34022	2.24813	2.34022
Bookkeeping Diff [mJ]	+0.00912	+0.01795	+0.08256	+0.01795
Bookkeeping Diff [%]	+0.3783%	+0.7607%	+3.5407%	+0.7607%
Status	TEMPORALLY SENSITIVE	Baseline Anchor	TEMPORALLY SENSITIVE	Baseline Anchor

7.8 Spatial Discretization Convergence of Global Energy Components (S_1 – S_4)

The spatial convergence of UEL energetic quantities is rigorously evaluated across the four authoritative governed fixed meshes: S_1 ($h = 3.00 \mu\text{m}$, 15,192 elements, Job 1406015), S_2 ($h = 2.00 \mu\text{m}$, 32,130 elements, Job 1406016), S_3 ($h = 1.50 \mu\text{m}$, 41,912 elements, Job 1406017), and S_4 ($h = 1.25 \mu\text{m}$, 51,408 elements, Job 1406018).

Table 9 and Figure 30 present the matched-displacement comparison at the three accepted physical checkpoints:

1. Pre-Peak Elastic and Damage Initiation ($u = 5.50 \mu\text{m}$):

- External boundary work W_{trap} is tightly invariant across meshes ($2.03739 \rightarrow 2.03528 \text{ mJ}$, -0.10%), classified **STABLE_OVER_TESTED_RANGE**.
- Stored elastic strain energy E_{elas} constitutes over 97.1% of total input work and varies by -0.23% ($1.98181 \rightarrow 1.97717 \text{ mJ}$), classified **STABLE_OVER_TESTED_RANGE**.
- Regularized fracture surface energy E_{frac} accounts for the remaining 2.9% of input work, increasing smoothly from 0.05572 mJ (S_1) to 0.05815 mJ (S_4 , $+4.36\%$), classified **MESH-SENSITIVE**.
- The two-term bookkeeping difference Δ_{book} is strictly negligible in this pre-peak regime (-0.00014 mJ to -0.00004 mJ , $< 0.007\%$ of W_{trap}).

2. Near-Peak / Early Softening Transition ($u = 5.85 \mu\text{m}$):

Reflects the mesh-induced peak displacement shift ($u_{\text{peak}} = 5.857 \mu\text{m}$ for S_1 vs $5.586 \mu\text{m}$ for S_4). Baseline S_1 resides directly at its load peak ($F = 0.75756 \text{ kN}$), whereas refined meshes S_2 – S_4 have already undergone rapid crack extension ($F < 0.0003 \text{ kN}$, $E_{\text{frac}} \approx 2.33 - 2.38 \text{ mJ}$).

3. Post-Peak Fully Fractured State ($u = 6.20 \mu\text{m}$):

- All four models have completely severed the ligament ($d_{\text{max}} \geq 1.0004$, residual load $F < 0.0006 \text{ kN}$, $E_{\text{elas}} < 0.0016 \text{ mJ} \approx 0$).
- Regularized fracture surface energy evaluated across the meshes gives $E_{\text{frac}} = 2.33886 \text{ mJ}$ (S_1), 2.33022 mJ (S_2 , -0.37%), 2.35701 mJ (S_3 , $+0.78\%$), and 2.37531 mJ (S_4 , $+1.56\%$). The full min–max spread across S_1 – S_4 is 1.94% , classified as **STABLE_OVER_TESTED_RANGE**.
- External boundary work W_{trap} decreases from 2.35801 mJ (S_1) to 2.17004 mJ (S_4 , -7.97%), classified **MESH-SENSITIVE**.
- The resulting post-peak two-term difference Δ_{book} ranges from $+0.01754 \text{ mJ}$ ($+0.74\%$ for S_1) to -0.20585 mJ (-9.49% for S_4), strictly designated as an observed balance residual under **GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED**.

Table 9: Spatial energy convergence audit across fixed meshes S_1 – S_4 at matched displacements.

Model (h)	Elements	$F(u)$ [kN]	W_{trap} [mJ]	E_{elas} [mJ]	E_{frac} [mJ]	E_{tot} [mJ]	Δ_{book} [mJ]	Classification
State 1: Pre-peak initiation ($u = 5.50 \mu\text{m}$)								
S_1 ($3.00 \mu\text{m}$)	15,192	0.72066	2.03739	1.98181	0.05572	2.03753	-0.00014	Baseline Qualified
S_2 ($2.00 \mu\text{m}$)	32,130	0.71984	2.03635	1.97957	0.05686	2.03643	-0.00008	STABLE (-0.05%)
S_3 ($1.50 \mu\text{m}$)	41,912	0.71925	2.03565	1.97795	0.05775	2.03570	-0.00005	STABLE (-0.09%)
S_4 ($1.25 \mu\text{m}$)	51,408	0.71897	2.03528	1.97717	0.05815	2.03532	-0.00004	STABLE (-0.10%)
State 2: Near-peak / softening transition ($u = 5.85 \mu\text{m}$)								
S_1 ($3.00 \mu\text{m}$)	15,192	0.75756	2.29636	2.21586	0.08068	2.29654	-0.00018	Peak State ($u_{\text{peak}} = 5.857 \mu\text{m}$)
S_2 ($2.00 \mu\text{m}$)	32,130	0.00031	2.24774	0.00090	2.33015	2.33105	-0.08331	Post-softening ($d_{\text{max}} = 1.0006$)
S_3 ($1.50 \mu\text{m}$)	41,912	0.00023	2.18985	0.00067	2.35696	2.35763	-0.16778	Post-softening ($d_{\text{max}} = 1.0007$)
S_4 ($1.25 \mu\text{m}$)	51,408	0.00020	2.16997	0.00057	2.37528	2.37585	-0.20588	Post-softening ($d_{\text{max}} = 1.0006$)
State 3: Post-peak fully fractured state ($u = 6.20 \mu\text{m}$)								
S_1 ($3.00 \mu\text{m}$)	15,192	0.00052	2.35801	0.00161	2.33886	2.34047	$+0.01754$	Broken Baseline ($d_{\text{max}} = 1.0004$)
S_2 ($2.00 \mu\text{m}$)	32,130	0.00028	2.24785	0.00087	2.33022	2.33110	-0.08325	E_{frac} STABLE (-0.37%)
S_3 ($1.50 \mu\text{m}$)	41,912	0.00021	2.18993	0.00066	2.35701	2.35767	-0.16774	E_{frac} STABLE ($+0.78\%$)
S_4 ($1.25 \mu\text{m}$)	51,408	0.00018	2.17004	0.00057	2.37531	2.37589	-0.20585	E_{frac} STABLE ($+1.56\%$)

7.9 Itemized Energetic Qualification Status Summary

Table 10 provides the definitive itemized status of all energetic expressions, separating verified source terms, derived relations, and unclosed physical questions.

Mode-I Primary Spatial Energy Convergence (S_1 -- S_4 , Single-IP Extraction Rule)

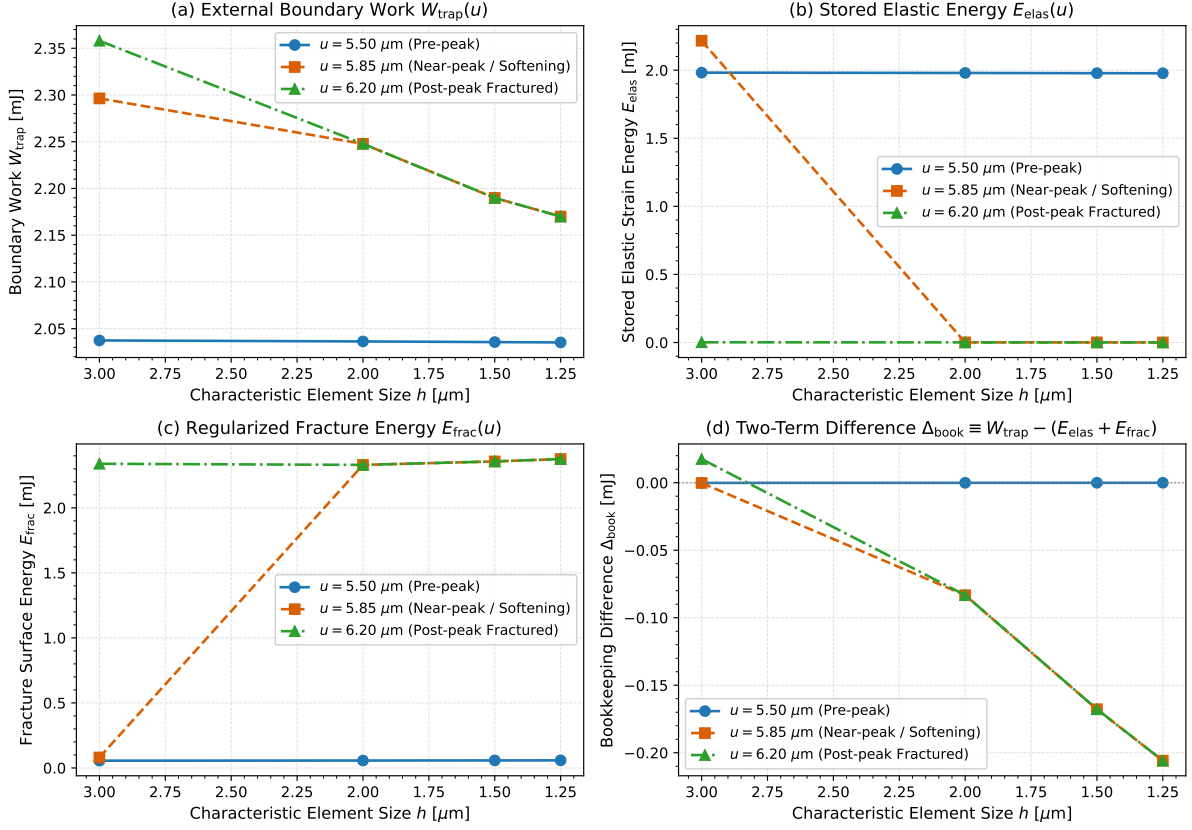


Figure 30: Spatial discretization convergence of global energy components across meshes S_1 – S_4 at matched displacements $u = 5.50, 5.85, 6.20 \mu\text{m}$: (a) boundary work W_{trap} ; (b) stored elastic strain energy E_{elas} ; (c) regularized fracture surface energy E_{frac} ; (d) two-term bookkeeping difference Δ_{book} . **GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED.**

Scientific conclusion. At $u = 5.50 \mu\text{m}$, boundary work and elastic energy are spatially stable. After fracture, E_{frac} is comparatively stable ($S_1 \rightarrow S_4$: +1.56%; full spread: 1.94%), while W_{trap} and Δ_{book} remain mesh-sensitive. A closed global energy identity has not been demonstrated.

Table 10: Definitive energetic quantity classification and qualification status.

Quantity	Mathematical Definition / Extraction	Units	Status
K_0	$dF/du _{\text{elastic}}$ (RP 999999 RF_2)	kN/mm	NUMERICALLY_VERIFIED
$F_{\text{max}}, u_{\text{peak}}$	Peak reaction force and displacement	kN, mm	NUMERICALLY_VERIFIED
W_{trap}	$\sum \frac{1}{2}(F_n + F_{n-1})(u_n - u_{n-1})$	kN · mm $\equiv$ J	DERIVED_AND_CHECKED
ψ_e	$\frac{1}{2}g(d)\epsilon : C_0 : \epsilon$	kN/mm ² = J/mm ³	VERIFIED_FROM_SOURCE
E_{elas}	$\int_{\Omega} \psi_e d\Omega \rightarrow \text{ENERGY}(2) \rightarrow \text{ALLSE}$	kN · mm $\equiv$ J	VERIFIED_FROM_SOURCE
ψ_f	$G_c[d^2/(2l_0) + (l_0/2) \nabla d ^2]$	kN/mm ² = J/mm ³	VERIFIED_FROM_SOURCE
E_{frac}	$\int_{\Omega} \psi_f d\Omega \rightarrow \text{STATEV}(17), \text{CSV}$	kN · mm $\equiv$ J	VERIFIED_FROM_SOURCE
Pre-peak Δ_{book}	$ \Delta_{\text{book}} /W_{\text{trap}} < 0.007\%$	Dimensionless	NUMERICALLY_VERIFIED
Post-peak Δ_{book}	$W_{\text{trap}} - (E_{\text{elas}} + E_{\text{frac}})$ residual (+0.7607% to -9.49%)	kN · mm $\equiv$ J	NUMERICALLY_VERIFIED (OBSERVED)
Mechanical Non-Invasiveness	Pointwise $F - u$ parity across 7,000 increments ($\Delta K_0 = 0.0\%$, $\Delta F_{\text{max}} = 0.0\%$)	—	QUALIFIED_MECHANICALLY_NONINVASIVE
$\mathcal{D}_{\text{frac}}$	$\int_0^t \int_{\Omega} 2(1-d)\mathcal{H}\dot{d} d\Omega d\tau$	kN · mm $\equiv$ J	NOT_YET_QUALIFIED
$\mathcal{D}_{\text{frac}} = \dot{E}_{\text{frac}}$	Rate equality under active damage	W	NOT_YET_QUALIFIED

8. Epistemological Audit, Claim Ledger, and Supervisory Decisions

8.1 Compact Supervisor-Facing Claim Ledger

Table 11 classifies the standing of all findings into three mutually exclusive categories: THEORY, VERIFIED NUMERICAL EVIDENCE, and UNRESOLVED.

Table 11: Compact Mode-I Claim Ledger across Core Physical and Energetic Metrics.

Metric / Aspect	THEORY (Continuum / Formulation)	VERIFIED EVIDENCE (Audited Artifacts)	UNRESOLVED (Boundaries / Open Scope)
1. Initial Stiffness K_0	Continuum linear elasticity ($\sigma = \mathbb{C}_0 : \epsilon$) provides reference stiffness.	$K_0 = 137.84\text{--}137.95\text{ kN/mm}$ ($\Delta < 0.09\%$ spatial across $S_1\text{--}S_4$, $\Delta < 0.001\%$ temporal across $T_1\text{--}T_3$). Baseline 14k adaptive: 137.909558 kN/mm vs $2\times$ Refined 137.909975 kN/mm ($\Delta K_0 = +0.000302\%$).	None for standard Mode-I; CONVERGED / STABLE .
2. Spatial Peak Force $F_{\max}$	Degradation $g(d) = (1 - d)^2 + k_{\text{res}}$ couples strain energy to damage evolution.	Monotonic reduction: $F_{\max} = 0.7578\text{ kN}$ (S_1) $\rightarrow$ 0.7312 kN (S_4) $\rightarrow$ 0.7255 kN (historical fine mesh), 4.26% drop across $h/l_0 \in [0.133, 0.400]$.	Asymptotic spatial convergence is not established within the tested mesh range; MESH-SENSITIVE .
3. Temporal Peak Force $F_{\max}$	Temporal invariance of limit load under Δt scaling.	Invariant: $F_{\max} = 0.75815\text{ kN}$ (T_1) $\rightarrow$ 0.75763 kN (T_3), spread $\Delta < 0.069\%$. On 14k adaptive mesh: $F_{\max} = 0.743701\text{ kN}$ (baseline) vs 0.743530 kN ($2\times$ refined), $\Delta F_{\max} = -0.0229\%$.	Pre-peak response and peak characteristics: CONVERGED / STABLE (QUALIFIED_OVER_PREPEAK_INTERVAL_ONLY).
4. Pre-Peak Work W_{trap}	Trapezoidal boundary-work estimate $W_{\text{trap},n} = \sum_{i=1}^n \frac{F_i + F_{i-1}}{2} (u_i - u_{i-1})$.	$W_{\text{trap}} \in [2.035, 2.038]\text{ mJ}$ at $u = 5.50\text{ }\mu\text{m}$ across tested cases ($\Delta < 0.15\%$). On 14k adaptive mesh, total energy at peak $u = 5.730\text{ }\mu\text{m}$ matches bitwise ($E_{\text{model}} = 2.205225\text{ mJ}$, $\Delta = 0.000000\text{ mJ}$).	Stable over tested pre-peak range; CONVERGED / STABLE .
5. Terminal Work W_{trap}	Endpoint integral along displacement path.	Full runs: $W_{\text{trap}} = 2.4101\text{ mJ}$ (T_1) $\rightarrow$ 2.3319 mJ (T_3) (3.24% variation); cutback runs truncated at $u = 6.8\text{--}9.6\text{ }\mu\text{m}$. $2\times$ refined adaptive run reaches $u = 7.470\text{ }\mu\text{m}$ (8,958 incs).	Post-peak cutback is sensitive to increment-scale displacement tolerance check ($c_{\max} \leq C_n \Delta u_{\text{inc}}$); TEMPORALLY SENSITIVE POSTPEAK .
6. Length-Scale (l_0) Sensitivity	Characteristic width of regularized damage band.	3-point fixed- S_3 study: $l_0 \in \{7.5, 11.25, 15.0\}\text{ }\mu\text{m}$ yields K_0 spread 0.13% (STABLE), $F_{\max}$ drop 5.83% ($0.732 \rightarrow 0.690\text{ kN}$) and u_{peak} shift ($5.633 \rightarrow 5.579\text{ }\mu\text{m}$).	Terminal energy across unequal endpoints: NOT_YET_QUALIFIED_UNEQUAL_ENDPOINTS .
7. UEL Energy Terms	Implemented integrals of E_{elas} and E_{frac} in <code>f42_mixed_uel.for</code> .	Unit-105 CSV and ODB IP1: $E_{\text{elas}} = 1.1608 \times 10^{-3}\text{ mJ}$, $E_{\text{frac}} = 2.34021\text{ mJ}$; matched-displacement spatial energy convergence is quantified for $S_1\text{--}S_4$.	Terminal cross-mesh energy comparison at unequal cutback endpoints is not qualified.
8. Bookkeeping Difference	Algebraic difference $W_{\text{trap}} - (E_{\text{elas}} + E_{\text{frac}})$.	Baseline S_1 : $+0.7607\%$ ($+1.795 \times 10^{-2}\text{ mJ}$); $T_1 \rightarrow T_3$: $+0.378\% \rightarrow +3.541\%$.	Exact decomposition requires within-increment path; TEMPORALLY SENSITIVE .
9. Global Energy Identity	Staggered solution scheme lacks joint discrete scalar potential.	Non-commutativity $\partial^2 \Pi / \partial \mathbf{u} \partial d \neq \partial^2 \Pi / \partial d \partial \mathbf{u}$; ODB does not record within-increment paths.	GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED .
10. Damage-Weighted Centroid Trajectory	Symmetric Mode-I crack extension along centerline $y = 0.500\text{ mm}$.	Centroid offset $ y_c - 0.5\text{ mm} \leq 3.10\text{ }\mu\text{m}$ across all meshes; bounded within single element dimension.	Pure Mode-I symmetry preserved; CONVERGED / STABLE .

8.2 Decisions and Direction Requested for 08 October Meeting

- Mode-I Foundations Approval:** Review and formal acceptance of the updated Mode-I thesis chapters (Chapters 1–3, 7–8) incorporating all verified mechanical, adaptive, and energetic findings.
- Supervisor Decision on Gate 6B Closure:** Supervisory ruling on accepting the endpoint energetic qualification and acknowledging the formal epistemic boundary **GLOBAL_ENERGY_IDENTITY -- NOT_YET_CLOSED** as the closed state of Gate 6B.
- Authorization to Proceed to Gate 6C:** Transition to Gate 6C (State Transfer Energy Preservation) upon formal closure of Gate 6B.
- Maintenance of Scope Holds:** Confirmation that Mode-II, multi-crack configurations, and Gate 7 visualization remain paused until Gate 6C state transfer is completed.

9. Job Ledger, Artifact Provenance, and References

9.1 Authoritative Job Ledger

Table 12 lists the key Abaqus simulations constituting the Mode-I evidence base, executed on the TU Bergakademie Freiberg HPC cluster (`mmaster02`, queue `normal_imfdkfq`).

Table 12: Authoritative HPC Job Ledger for the Mode-I Benchmark Series.

Job ID	Model Name	Description	Input Deck Hash	Elements	Exit Status
1406015.mmaster02	PK_M1_S1_H0030	Canonical spatial baseline S_1 ($h = 3.0 \mu\text{m}$)	bc000e0c579...	15,192	Exit_status=0
1398090.mmaster02	PK_M1_FIX_REF	Historical reference anchor	a845ef1a37c...	15,192	Exit_status=0
1404933.mmaster02	PK_M1_NOM1_1404933	Corrected 71k full fracture	72674e2ba05...	71,320	Exit_status=0
1405044.mmaster02	PK_M1_FROZEN_71K	Frozen elastic requalification	3bbdfb9a528...	71,320	Exit_status=0
1406016.mmaster02	PK_M1_S2_H0020	Spatial fine S_2 ($h = 2.0 \mu\text{m}$)	34ec7f0b5d0...	32,130	Exit_status=0 (Cutback)
1406017.mmaster02	PK_M1_S3_H0015	Spatial fine S_3 ($h = 1.5 \mu\text{m}$, $l_0 = 7.5 \mu\text{m}$)	eb009f188a2...	41,912	Exit_status=0 (Cutback)
1406018.mmaster02	PK_M1_S4_H00125	Spatial fine S_4 ($h = 1.25 \mu\text{m}$)	cc88df1b08...	51,408	Exit_status=0 (Cutback)
1406019.mmaster02	PK_M1_S5_H00100	Historical fine ($h = 1.0 \mu\text{m}$; legacy model name; not governed S_5)	634b3d758f8...	69,384	Exit_status=0 (Cutback)
1406020.mmaster02	PK_M1_T1_DT2X	Temporal coarse T_1 ($2.0 \times \Delta t$)	ad9dfa6d899...	15,192	Exit_status=0
1406021.mmaster02	PK_M1_T2_DT1X	Temporal nominal T_2 ($1.0 \times \Delta t$)	fa88301c272...	15,192	Exit_status=0
1406317.mmaster02	PK_M1_T3_DT05X_R2	Temporal fine T_3 ($0.5 \times \Delta t$)	92562479633...	15,192	Exit_status=0 (Full)
1406278.mmaster02	PK_M1_A2_2PCT	Adaptive A_2 (2% errorTarget)	d4ec8aa9ebf...	15,396	Exit_status=0
1406273.mmaster02	PK_M1_A3_3PCT	Adaptive A_3 (3% errorTarget)	bb185fca481...	7,633	Exit_status=0
1406311.mmaster02	PK_M1_A3_FINE_DT	Adaptive A_3 fine Δt ($0.5 \times$)	8c14cf29b4e...	7,633	Exit_status=0
1406279.mmaster02	PK_M1_A4_5PCT	Adaptive A_4 (5% errorTarget)	82bcf2ff1c5...	4,194	Exit_status=0
1406839.mmaster02	PK_M1_S1_DIAG_R2	Historical diagnostic energy-output run; non-authoritative for canonical S_1 provenance	fa88301c272...	15,192	Exit_status=0
1406895.mmaster02	PK_M1_S3_L01125	$l_0 = 11.25 \mu\text{m}$ on fixed S_3 mesh	77df64d10d0...	41,912	Exit_status=0 (Full)
1406896.mmaster02	PK_M1_S3_L01500	$l_0 = 15.00 \mu\text{m}$ on fixed S_3 mesh	ff13ba60800...	41,912	Exit_status=0 (Full)

9.2 Subroutine Integrity

The authoritative user subroutine `f42_mixed_uel.for` governing all production calculations has the immutable cryptographic hash:

SHA256: 5cd0d2c015c9ead91c99d7a744156cc86f5b5ea26473bbbed7d6e5515fe30fa46

References

Abaqus. *Abaqus 2023 Documentation: Analysis User's Guide, User Subroutines Reference Guide, and CAE User's Guide*. Dassault Systèmes Simulia Corp., Providence, RI, USA, 2023.

A. Pandey and S. Kumar. A simple and robust mesh refinement implementation in Abaqus for phase field modelling of brittle fracture. *Computer Modeling in Engineering & Sciences*, 144(3):3251–3286, 2025. doi: 10.32604/cmes.2025.067858.